

The Structure of Admissibility

Bivalent Qualification, Exhaustion, and the Unique Admissible Interior

Amos Jay Maley

Publication reference edition, 21 September 2026

Abstract

Non-degenerate determinate construction necessarily instantiates admissibility, standing, reference, and irreversibility. This paper establishes the structure of the governed interface through three connected results. First, the complete observation profile has a canonical minimal quotient, which reduces to one qualification bit exactly when every required observation, including application-definedness, is constant on that bit's fibres. Second, an independently specified reuse contract determines a unique greatest safe interior. It contains every sound invariant candidate, and every excluded configuration has a finite failed-qualification or failed-warrant witness. Third, an exhaustion theorem connects these exact classifications to the possible failures of claimed complete realization.

The exhaustion starts from original result relations and intended uses fixed independently of the proposed implementation. A claimed complete graph or interior can fail through surplus, omission, or both. Complete observation semantics supplies the coverage bridge; the five failure loci distinguish illicit redescription, incorrect qualification or unwarranted entry, historical substitution, unsupported scope transfer, and unlicensed decision structure. Their causes may overlap. Each exclusion follows from a specific semantic or warrant failure.

The local AMetric and fibre tests establish the associated impossibility results: a description cannot recover distinctions it has discarded, an unmarked input cannot warrant selection of an atom from a non-singleton equality-only carrier, and an unseeded rule closure cannot supply its own authorization. Full supporting boundary proofs appear in an appendix. Original-incidence conservation and retained-history non-repair are distinguished from negative endpoint absorption. Finite protocol and typed-calculus examples show both successful realizations and the failures exposed by the tests. The result is a constructive and exhaustive account of admissibility for the complete determinate-use problem, with its mathematical, semantic, and coverage obligations explicit.

Keywords: admissibility; standing; objecthood; invariance; bivalence; observational equivalence; exhaustion; greatest fixed point; conservation.

Contents

1	Introduction: what an admissibility structure must retain	4
1.1	A familiar example: an unsupported bridge trial	4
1.2	The main result, with its objects distinguished	5
1.3	Dependence map and four decisive tests	5
1.4	Literature and division of labor	6
1.5	What would count against the claims	7
2	Determinate incidence, kernel necessity, and reference	7
2.1	Complete questions before their answers	7
2.2	Reference as a complete invariant	8
2.3	The distinction that conservation needs	9
3	The local AMetric constraints used by the structure theorem	9
3.1	Specify the content before invoking symmetry	10
3.2	The elementary local obstruction	10
3.3	From the boundary to the interior	10
4	Non-transmission, grounded authorization, and faithful extension	11
4.1	The fibre test	11
4.2	Truth, evidence, and permission to use a certificate	12
4.3	Faithful extensions and same-scope closure	13
5	Qualification, complete observations, and binary reduction	14
5.1	A binary qualification and its semantic scope	14
5.2	The canonical interface for a complete use family	16
5.3	Primitive checks that generate a complete continuation interface	17
6	Standing as the greatest invariant safe interior	19
6.1	Invariance and faithful transport of the interior	22
7	Conservation, new constructions, and retained history	23
8	Semantic exhaustion and impossibility of same-scope evasion	24
8.1	Original result graphs: surplus, omission, and negative claims	24
8.2	Information and interiors: exact alternatives with witnesses	25
8.3	Coverage before the five mechanism names	26

8.4	Impossibility dispositions and the local gate	30
9	Worked realizations and discriminating examples	31
9.1	A finite bridge protocol	31
9.2	The finite data actually needed for a successful bit interface	32
9.3	Simply typed construction: an independently justified interior	33
9.4	An original answer can be unknown and still conserved	34
10	The canonical architecture and its downstream use	35
10.1	The complete quotient preserves the complete interior	35
10.2	One theorem chain, with distinct sources of necessity	36
10.3	Mathematics and physics	37
10.4	Assessment of the theorem’s force	37
A	Claim map and proof dependencies	38
A.1	Acyclic proof order	39
A.2	Logical and computational conventions	39
B	Notation, scope, and verification sequence	40
B.1	Four quantities that must not be identified	40
B.2	Recommended verification sequence	41
B.3	Formalization boundary	41
C	Supporting AMetric classification and selection proofs	41
C.1	From absence of an individuator to the right symmetry	42
C.2	Complete finite-arity classification	42
C.3	The stabilizer test and what selection requires	43
C.4	Metrics, probabilities, and typicality	44
C.5	The local boundary conclusion	45
	References	45

1 Introduction: what an admissibility structure must retain

A determinate construction has something that counts as its subject, conditions under which a proposed use is justified, and a difference between carrying that subject forward and substituting another. These commitments occur in mathematics and physics alike. A derivation must license its conclusion from the premises actually available. A record must concern the occurrence it purports to record. A physical endpoint must be the endpoint of the specified incidence under the specified conditions. Admissibility is the work that holds these requirements together in constructional use.

The kernel theorem identifies four necessary functional roles: Admissibility, Standing, Reference, and Irreversibility [8]. Their application does not await a choice to interpret an independently complete physical domain through an additional formalism. Establishing a non-degenerate determinate object, state, carrier, record, or endpoint already incurs the work. What must still be established in a particular domain is its substantive qualification condition, the relevant comparison class, and which continuations preserve which answers. Necessary governance does not supply these domain facts merely by naming the roles.

This paper supplies the next stage: a precise account of the structure of the governed interface. Its central results are the exact binary-reduction criterion (Theorem 5.6), the greatest safe interior (Theorem 6.3), and failure exhaustion against the independently specified original obligations (Theorem 8.6). It separates three questions often compressed into the word “status.” Does this occurrence qualify for this complete use-question? What complete information about qualification and continuation must be retained? Which qualified states remain eligible throughout the specified uses? The answers are, respectively, a bit, a canonical profile, and a greatest stable interior. Each has its own universal characterization.

1.1 A familiar example: an unsupported bridge trial

Consider a small model bridge tested with a stated load. The question is whether it supported that load without outside support during a particular trial. Fix the bridge, load, span, support conditions, and trial occurrence. A person secretly holding up the bridge does not supply a successful unsupported trial. Repairing the bridge and repeating the test may supply a successful second trial. Discovering a recording of the original test may correct an earlier report without changing what occurred.

Each kernel role has visible work here. Reference keeps the report about the specified bridge and trial. Standing is the trial’s eligibility to support the stated performance claim. Admissibility excludes a substituted load, outside support, or a result belonging to another trial. Irreversibility retains the original incidence when a later repair or discovery occurs. Deleting the terminology leaves this work intact; deleting the work loses the corresponding claim.

Two bridges can both pass the first trial and differ in whether they survive a second load. Their first qualification answers agree. Their continuation profiles differ. A full engineering record must retain that difference if the second load is among the promised uses. Conversely, two labels for the same complete record cannot properly yield discordant answers to the same complete question. This is the distinction between legitimate post-fixational structure and a merely presentational difference.

1.2 The main result, with its objects distinguished

Theorem 1.1 (Structure of admissibility: statement of the proof program). *For a non-degenerate determinate construction domain, fix complete questions, their correct qualification relation, and an independently declared family of relevant continuations and observations. Then the following conclusions hold under the explicit local hypotheses proved below.*

- (i) *Each complete evaluated qualification question is bivalent. The complete observation profile has a unique minimal lossless interface up to bijection. That interface reduces to one qualification bit exactly when all its required observations factor through that bit.*
- (ii) *For a fixed reuse contract, reuse-stable qualification has a greatest interior, characterized by all finite safe paths and by a greatest fixed point. Every sound closed candidate is contained in it; completeness identifies it uniquely. Every excluded configuration has a finite local or edge-warrant failure witness.*
- (iii) *A claimed complete result graph or interior either agrees with its original target, includes an unsupported result or state, or omits a required one; the two failures may coexist. With independently established coverage of the original questions and uses, these witnesses and their warrant dependencies yield the five-locus exhaustion. The associated impossibility arguments exclude lossless recovery from insufficient content, discordant warranted answers at one original locus, and unseeded authorization.*
- (iv) *Faithful representations preserve original-incidence answers. Retained failures cannot be erased from the history they concern. Negative endpoint absorption is a separate property, characterized by whether endpoint qualification certifies all earlier qualifications on the specified paths.*
- (v) *Every established determinate use necessarily instantiates the four kernel roles. A complete reference quotient represents identity. The local AMetric and fibre constraints state what the advertised content can determine; their full supporting proofs appear in Appendix C.*

The following sections prove each part of this statement. The observation family in (i), the reuse contract in (ii), and the original-obligation coverage in (iii) are explicit mathematical data. Coverage must be justified from the original task, not stipulated by declaring a preferred list exhaustive. A complete positive use necessarily performs the kernel work. The question whether an encoding, symmetry, or transition is adequate is settled by its properties, not by calling it admissible in advance.

1.3 Dependence map and four decisive tests

The dependence runs through eight steps. The main structural conclusions lead the exposition above; the proofs establish their premises in the following order:

1. Fix the original incidence and the complete question independently of its verdict; recover the reference quotient.
2. Identify the full pre-fixational content and its actual automorphisms; classify its invariant predicates and available selectors.
3. Apply the fibre test to the proposed transfer from that content to a determinate answer; locate the substantive warrant in any successful construction.

4. Distinguish binary qualification from the total observation profile; construct the canonical interface and prove the exact criterion for binary reduction.
5. Declare relevant reuse independently of the desired standing set; construct the greatest stable interior and prove completeness and uniqueness.
6. Prove conservation for the original incidence and retention for histories; state the separate endpoint criterion for negative absorption.
7. Prove failure exhaustion from original result graphs, complete observations, use warrants, and finite-path witnesses; locate the five failure mechanisms and their impossibility consequences.
8. Verify the resulting obligations in concrete mathematical and physical examples.

Four mathematical tests recur throughout. The *fibre test* asks whether forgotten distinctions can change the required answer. The *stabilizer test* asks whether the available structure fixes an eligible selected output. The *lossless-interface test* asks whether all relevant observations can be recovered. The *interior test* asks whether the candidate is qualified, closed under the declared warranted uses, and complete among such candidates. These tests assess a proposed realization against its independently stated mathematical requirements.

Exhaustion and impossibility have complementary roles. Exhaustion proves that the stated alternatives cover the original problem. An impossibility proof then closes a defective alternative. Construction of the greatest interior identifies the complete surviving region. Neither an inventory of named failures nor an exclusion proof alone supplies all three steps.

The pre-metric argument controls what can enter an evaluation without additional structure; it does not collapse distinctions after substantive structure has been supplied. Bivalence concerns each complete qualification question; it does not assert that all qualified objects have the same behavior. Uniqueness concerns the complete admissible interior for one fixed contract; it does not identify all its elements or all possible domain contracts.

1.4 Literature and division of labor

The foundational use of identity and persistence is adjacent to Wiggins’s treatment of sameness [1]. The move from an asserted target capacity to its necessary conditions also faces the challenge associated with transcendental arguments: the conditions must be defended as requirements of the target, rather than slipped into an unexplained vocabulary [2]. Section 2 therefore states the intended work and its deletion failures. The kernel paper carries the full reconstruction; this paper makes only the local commitments needed by the interface results.

The mathematics uses established constructions. Names and equivariance belong to the setting studied by nominal-set theory [3]. Observational identification and systems organized by their observable behavior are close to the coalgebraic perspective [4]. Greatest fixed points and the order of invariant subsets belong to the fixed-point tradition [5]. Typed operational examples use the familiar distinction between preservation and its generally invalid converse [6, 7]. These connections locate the methods; they do not supply unproved premises about a particular physical interpretation.

The contribution is their assembly into an explicit admissibility structure, including the original-incidence interpretation, exact binary-reduction criterion, complete interior, failure exhaustion, and separate conservation claims. The elementary quotient and fixed-point facts are not claimed as new general mathematics. Their premises are exposed so that a domain application cannot import a

favorable conclusion by silently choosing its observations or continuations.

The AMetric reference manuscript develops the boundary analysis more broadly [9]. The local arguments needed here appear in Section 3 and Appendix C; no consequence that depends on the present structure theorem is imported as a premise. Other corpus manuscripts provide applications and comparisons. The proof dependencies established here are listed in Appendix A.

1.5 What would count against the claims

A successful implementation under different terminology instantiates the corresponding kernel work. A challenge to constitutive necessity must preserve the claimed determinate use while actually removing that work. The mathematical claims have more direct tests: exhibit a target answer differing within an alleged sufficient fibre; an alleged selector moved by the stabilizer; a relevant observation lost by the proposed interface; or a candidate interior that omits an eligible stable component while claiming completeness. An exhaustion claim can be defeated by an original required use absent from its coverage argument, or by a claimed failure disposition that is not entailed by its witness. Two warranted answers that disagree about the same original incidence and complete question would defeat the conservation claim. An endpoint that qualifies after an earlier unqualified endpoint defeats negative absorption for that transition system, while leaving historical non-substitution untouched.

These tests locate failures in the domain data, interface completeness, invariance argument, or derivation.

2 Determinate incidence, kernel necessity, and reference

2.1 Complete questions before their answers

Definition 2.1 (Assessment contract). An assessment contract D specifies a carrier of candidate occurrences or configurations X , their target identifications, the substantive question about each occurrence, its original conditions, and the observations and uses whose preservation is at issue. A complete evaluated query fixes these data sufficiently to settle the qualification relation being asserted. Candidate representations and continuations can be declared before their fidelity or success is established. Occurrence identity is not defined by the verdict.

For a mathematical judgment, an occurrence can be an instantiation of a rule or a position in a derivation. For a physical trial, it can be the trial under its actual conditions. Different inscriptions can represent one occurrence. Two occurrences can receive the same verdict. Neither fact confuses numerical identity with qualification.

The contract terminology does not make physical constraints conventional. When the question concerns an actual incidence, its substantive relations supply the conditions; a description is accountable to them. A stipulated formal calculus supplies a different kind of condition. The common structure is that the offered answer concerns this target and these conditions. A mistaken choice of physical conditions is a mistaken model, not a change in the physical incidence's governance.

Definition 2.2 (The four functional roles). *Reference* identifies what the use concerns. *Standing* is eligibility for that specified use under its actual conditions. *Irreversibility* retains the original incidence and conditions against replacement by a later correction. *Admissibility* governs entry into

constructional reliance with these requirements respected. These roles concern work performed by a realization, not four independent pieces of machinery or four primitive logical axioms.

Proposition 2.3 (Necessary governance and functional deletion). *Every successful non-degenerate determinate constructional use incurs all four roles. A package that functionally removes any role ceases to realize the corresponding complete use. The same conclusion applies to determinate physical objects, states, carriers, records, and endpoints; no further choice to adopt the kernel is needed.*

Proof. A purported determinate use must concern its identified bearer. Removing the work that keeps the assessment about that bearer removes reference and loses determinate attribution. It must distinguish what the bearer actually qualifies for from what would be true only of another bearer or under different conditions. Removing that eligibility distinction loses standing for the specified use. A claim about an original occurrence must remain about that occurrence under its original conditions. Substituting a repaired performance while retaining the original claim loses the fixation expressed by irreversibility. Finally, combining or relying on these attributions must respect all these conditions. Removing that control permits unsupported reliance or substitution, which loses admissibility. An alternative implementation that still performs the work has not functionally removed it. These are the local deletion arguments underlying the fuller kernel reconstruction [8]. An unobserved physical incidence already has the relata and original conditions in virtue of which some identity-bearing attributions are correct and substituted attributions are not. Those distinctions belong to its actual participation in the relation; they are not created by a later observer or checker. The same role reconstruction therefore applies to that incidence before any report about it. $\square$

An unsuccessful bridge trial remains a determinate physical occurrence. Its failure removes its standing for the claim of successful unsupported performance; it does not remove the objecthood of the bridge or the trial. An inaccurate record can also be a perfectly determinate physical object. It lacks the evidential relation claimed of it. The roles govern each of these determinate incidences necessarily, while their particular qualification answers differ.

The proposition is a constitutive argument about determinate use. It is not a deduction of physical laws from membership in an uninterpreted set. Nor does it derive a result for every syntactically expressible question. Where a question is incomplete or its semantics supplies several possible outcomes, that limitation must be represented. The bivalence result later concerns the complete qualification relation, not an imposed choice of a physical outcome or a decision algorithm.

2.2 Reference as a complete invariant

Let E be the independently justified equivalence relation on representations in X expressing identity for a specified reference purpose. If the starting data are reversible identifications, take the equivalence relation they generate. If they are arbitrary directed transitions, their connected components need not be identity classes: a transition can change the state or the target property. The proof therefore requires a reference relation, not merely a path relation.

Theorem 2.4 (Complete reference invariant). *The quotient map $q : X \rightarrow X/E$ is constant on E -classes and separates distinct classes. Every map $f : X \rightarrow Z$ constant on E -classes factors uniquely as $f = \tilde{f} \circ q$. An invariant separating at least two reference classes exists exactly when X/E has at least two elements.*

Proof. Define $q(x) = [x]_E$. Equality $q(x) = q(y)$ is equivalent to xEy , giving both assertions about identification. Define $f([x]_E) = f(x)$. Constancy makes this well-defined; surjectivity of q gives uniqueness. If there are two classes, q itself distinguishes them. If there is at most one class, every class-constant map has image of cardinality at most one. $\square$

This proof avoids a vacuous criterion. A constant function is invariant under every transformation, so the mere existence of an invariant says nothing about successful reference separation. The quotient provides the exact separation claim. Its relevance to actual reference depends on the independent justification of E . Declaring two physically different occurrences equivalent cannot make the quotient a faithful account of their identity.

Proposition 2.5 (Faithful recoding of reference). *Suppose $e : X \rightarrow Y$ and equivalence relations E on X , F on Y satisfy*

$$xE x' \iff e(x)Fe(x').$$

Then $[x]_E \mapsto [e(x)]_F$ is an injection from X/E into Y/F . It is a bijection exactly when every F -class has an e -representative. Every reference invariant can then be transported without loss.

Proof. The forward implication establishes well-definedness; the reverse establishes injectivity. The stated coverage condition is precisely surjectivity. Factoring through the quotient and then through this bijection transports each invariant. $\square$

2.3 The distinction that conservation needs

A redescription preserves a specified original fact when it represents that fact with the appropriate translation. A dynamical evolution may instead connect two distinct configurations of one persisting object. Persistence of the object does not make every predicate of its configurations invariant. “The bridge is intact” can change while the bridge remains the subject of a history. In the same way, a mathematical rewrite can preserve denotation while changing syntax or a local proof obligation.

Irreversibility in this paper is original-incidence fixation. No assumption that every physical process is noninvertible is required. Conversely, the existence of one nonreturning transition in an abstract graph would not establish historical fixation, binary status, or absorption of all negative states. Those are separate properties and receive separate proofs.

The positive existence condition is also local. Non-degenerate standing-bearing use supplies some actual successful instance. It need not supply both positive and negative examples for every fixed question. The exact two-class theorem therefore states the additional two-sided inhabitation condition where it is needed. This prevents the proof from creating a negative class, or a positive one, solely by notation.

3 The local AMetric constraints used by the structure theorem

The local boundary specifies what an advertised input can determine. This section states the content and symmetry obligations; Section 4 gives the exact transmission test. Appendix C supplies the full equality-pattern, selection, metric, and probability proofs. The AMetric reference manuscript develops their wider pre-fixational setting [9].

3.1 Specify the content before invoking symmetry

Licensed proposal content comprises all the structure that may bear on the stated target use before the proposed fixation. An equality-only carrier has equality but no target-bearing marked atom, order, coordinate, adjacency, or weight. Its automorphism group is the full permutation group. When licensed relations are present, the actual automorphism group must preserve them. These are substantive content claims, not consequences of calling a variable dimensionless.

In particular, absence of a licensed atom selector does not imply absence of relations. An unmarked cycle has adjacency even when its symmetries fix no vertex. A determinate proposal record is itself necessarily kernel-governed; the question is whether it supplies what is needed for a different proposed target use. The boundary does not posit an ungoverned physical substrate.

A construction determined solely by licensed content must respect changes of presentation that preserve that content. Scalar answers must be invariant; structured outputs must be equivariant under the independently justified output interpretation. Proposition C.2 gives the accountability argument. A non-equivariant set-theoretic function does not become warranted by the input merely because the function exists.

3.2 The elementary local obstruction

Lemma 3.1 (Equality-only discrimination and selection). *On a fixed equality-only carrier P , two finite tuples have the same permutation orbit exactly when they have the same equality pattern. Hence every invariant finite-arity predicate is a union of equality-pattern classes. A unary invariant predicate on a nonempty P is constant. If $|P| \geq 2$, no deterministic equivariant construction selects an atom from an unmarked input fixed by every permutation.*

Proof. A permutation preserves the equalities between coordinates. Conversely, equal patterns define a bijection between the finitely many tuple entries; extending it to a permutation of their union and fixing all other atoms sends one tuple to the other. Invariant predicates are constant on these orbits. There is one unary orbit. An output selected from an input fixed by every permutation would have to be an atom fixed by every permutation, whereas a transposition moves any chosen atom when $|P| \geq 2$. $\square$

The full definability form is Theorem C.4. For richer inputs, Theorem C.6 replaces the unmarked-input obstruction by an exact eligible-fixed-point criterion. These statements support the *stabilizer test*: the symmetries fixing the advertised input must fix an eligible output. Additional licensed structure may satisfy that test. It then explains the successful selection instead of disappearing from its dependencies.

The restriction is to unsupported discrimination. The empty strict ranking, constant scores, discrete atom metrics, and finite uniform probability laws are genuine mathematical survivors. None selects a distinguished unmarked atom merely by existing. Appendix C proves these distinctions and characterizes the resulting implementations.

3.3 From the boundary to the interior

The mathematical obstruction needed downstream is sharper than a list of forbidden techniques: if a proposed description identifies situations whose required answers differ, no decoder of that

description alone can answer them both correctly. The fibre theorem below proves this for arbitrary maps, independently of whether the descriptions use numbers, symbols, probability distributions, or learned representations. A licensed added channel can remove the obstruction by retaining the missing distinction.

The argument therefore proceeds from the specified input to its recoverable answers, from those answers to a complete interface, and from the declared uses to their safe interior. The equality-only special case does not erase structure supplied in an evaluated construction. Exhaustion in Section 8 combines these obstructions with the independently specified original obligations; it does not assume that every evaluated interface is itself equality-only.

4 Non-transmission, grounded authorization, and faithful extension

4.1 The fibre test

Let E be a set of complete candidate situations, $a : E \rightarrow Z$ a required target answer, and $p : E \rightarrow P$ the information retained by a proposed lower description. A fibre is the collection of situations with one retained value: $p^{-1}(u)$. The description may discard a parameter, a historical occurrence, an operation domain, or an entire family of observations.

Theorem 4.1 (Exact transmission criterion). *There exists a unique map $\bar{a} : p(E) \rightarrow Z$ satisfying $a = \bar{a} \circ p$ exactly when*

$$p(e) = p(e') \implies a(e) = a(e') \quad (e, e' \in E).$$

Thus no procedure receiving only $p(e)$ can correctly recover $a(e)$ on all of E if one fibre contains discordant target answers.

Proof. A factorization gives the implication by substitution. Conversely define $\bar{a}(u)$ as the common value of a on the nonempty fibre above $u \in p(E)$. The implication makes that value unique, without a choice of representative. Surjectivity onto the image gives uniqueness of $\bar{a}$. A proposed universally correct procedure would provide such a factorization and is therefore excluded by any discordant fibre. $\square$

The theorem quantifies over complete candidate situations compatible with the advertised description. It does not assert two correct answers under one complete fixed question. For the bridge, a description retaining only “the deck stayed up” can identify an unsupported trial and an externally supported trial. If unsupported performance is the question, those cases have different qualification answers. No recoding of that short description alone resolves the difference.

Corollary 4.2 (Compositional non-transmission). *If a fails the fibre test for p , it also fails for every map $r \circ p$. Adding a second retained channel $s : E \rightarrow T$ recovers a exactly when a is constant on the joint fibres of (p, s) . Finer information can therefore remove a particular obstruction; a mere postprocessing of the original channel cannot.*

Proof. A discordant pair with equal p -values remains equal after r . The second assertion is Theorem 4.1 applied to the product map. $\square$

This is the pre-metric non-transmissive boundary in its general form. It covers arbitrary maps and arbitrary set-sized question families; no finite list of implementation styles is needed. In the equality-only case, Theorem C.4 determines exactly which proposal answers can be constant on the relevant orbits. The stabilizer test supplies the additional condition when the requested answer is an equivariant selected object rather than a scalar class value.

4.2 Truth, evidence, and permission to use a certificate

A correct answer can be discovered after the occurrence it concerns. A verifier can compute a predicate specified independently of the verifier. A proof can derive a sufficient condition for qualification. None of these activities becomes circular merely because the certificate was produced later. The issue is what justifies its application to the original complete question.

Definition 4.3 (Grounded certificate system). Fix the intended semantics of the conclusions, a set of warranted premises, and inference rules whose instances have independently established semantic soundness. A grounded certificate is a well-founded proof tree formed from these premises and rule instances. Each node states its interpreted conclusion, including the bearer and conditions when the conclusion concerns a target use. Finite dependency graphs are a special case; infinitary branching is allowed when its inference rule is sound and the tree is well-founded.

Theorem 4.4 (Grounded authorization closure). *Every conclusion of a grounded certificate is warranted under its stated interpretation. If a conclusion warrants a target-domain use, the complete justification either contains a warranted target-use premise or contains a rule instance that introduces such warrant. A system with neither kind of contribution cannot justify that use merely by emitting its assertion.*

Proof. Well-founded induction on the proof tree proves the first assertion: warranted leaves supply the basis, and the soundness of a rule supplies its conclusion from its warranted premises. For the second, consider the nonempty set of nodes whose interpreted conclusions warrant a target use. Well-foundedness gives a node in this set with no such predecessor in the dependency order. It is either a warranted premise, or its rule application supplies target-use warrant without a target-use conclusion among its dependencies. These are the stated alternatives. Omitting both removes every possible first contribution of the asserted warrant. $\square$

This first-authority argument does not assert that every mathematical proof uses a particular logical cut rule. It concerns semantic support. A syntactically cut-free derivation still has premises and sound inference steps. Cyclic certificates require a separate acceptance theorem, such as a justified induction or coinduction rule; the cycle by itself is not a well-founded proof. The fixed-point construction later supplies such a coinductive rule by direct proof, rather than treating a circular certificate as self-validating.

The theorem also does not make elementary output computations “authority-free” when their independently established meaning already warrants the target use. If a rule’s soundness proves that a measured relation entails the required qualification, its application performs the necessary admissibility work. Deriving that implementation is a real explanatory achievement. It is not functional deletion of the role.

Proposition 4.5 (Sufficient certificates need not equal the gate). *Let $A \subseteq E$ be the target qualification set and $C \subseteq E$ a certificate condition. If $C \subseteq A$ is established, every verified C -instance*

warrants its A -qualification even when $C \neq A$. If $A \subseteq C$ also holds, the certificate is complete as well as sound.

Proof. The first assertion is implication under the proved inclusion. The second is equality by the two inclusions. Neither argument requires an extra chronological evaluation event. $\square$

For example, a proof that a bridge supported a heavier specified load under conditions that entail support of the lighter load can supply a sufficient certificate, provided that implication is actually justified in the model. Such a certificate need not cover every bridge able to support the lighter load. Treating soundness as if it required extensional equality would wrongly exclude ordinary proof methods.

4.3 Faithful extensions and same-scope closure

Definition 4.6 (Faithful semantic extension). An extension of a fixed assessment contract is faithful for its original questions when its new language, encodings, proofs, or observations interpret the original bearer and conditions without substitution and return only their correct original answers. It may settle previously unknown answers, provide new certificates, or add different questions. Its fidelity is a property to be established, not an authorization obtained from the word “extension.”

Theorem 4.7 (Faithful-extension conservation). *A faithful extension cannot change the correct answer to an unchanged original complete question. It can enlarge the set of certified answers without changing their correctness conditions. If the extension computes a target answer from a reduct, its success requires the fibre criterion for that reduct; if it adds distinguishing information, success is assessed on the enlarged input. A correct sufficient certificate need only imply the original qualification condition.*

Proof. By fidelity each answer concerns the same original data and equals the answer fixed by their semantics. Two such answers cannot disagree. The set of available certificates can nevertheless grow, since correctness and discovery are different predicates. The reduct claim is Theorem 4.1; the enlarged-input claim is Corollary 4.2; the final assertion is Proposition 4.5. $\square$

An exhaustive test of a proposed extension examines its dependencies: the bearer and question; the information retained; the semantic correctness of the claimed answer; and the premises or rules supplying its warrant. Failures at these locations can overlap. An extension might both discard necessary data and substitute a different question. The resulting failure mechanisms can overlap.

Theorem 4.8 (No unseeded authorization by rule closure). *Let J be a set of interpreted judgments, $B \subseteq J$ the independently grounded premises, and $\mathcal{R}$ a set of rules (Γ, j) with $\Gamma \subseteq J$. Let $\text{Cl}_{\mathcal{R}}(B)$ be the intersection of all rule-closed supersets of B . For $C \subseteq J$, suppose $B \cap C = \emptyset$ and every rule concluding in C has at least one premise in C . Then*

$$\text{Cl}_{\mathcal{R}}(B) \cap C = \emptyset.$$

This holds for arbitrary premise sets and does not require a finite list of possible rule forms.

Proof. The set $J \setminus C$ contains B . It is rule-closed: a rule all of whose premises lie outside C cannot conclude in C , by the hypothesis on rules. The least closed superset of B is therefore contained in $J \setminus C$. The least closure exists because J is a closed superset and intersections of closed supersets are closed. $\square$

The hypothesis concerns every rule available to establish a judgment in C . A cyclic-looking collection of dependencies can coexist with an independent grounding route; in that case the hypothesis fails at the grounding route. The theorem does not confuse one ungrounded attempted proof with the absence of all other proofs. Nor does escape from the obstruction prove a proposed rule sound. The independent interpretation and justification of any new entrance remain necessary.

Corollary 4.9 (No governance-free faithful authorization). *A construction that successfully authorizes an original determinate use necessarily realizes the relevant kernel work, whether that work is implemented internally, derived, distributed, or supplied through a faithful external encoding. Redescription of the implementation cannot remove its target accountability. Where the advertised input fails the fibre test, additional warrant without additional distinguishing content cannot make it a universally correct decoder of the missing answer.*

Proof. Successful authorization is a determinate constructional use, so Proposition 2.3 applies. Theorem 4.4 locates its warrant when supplied by a grounded certificate. The impossibility of decoding the missing answer follows independently from Theorem 4.1. Hence the constitutive claim and the information-theoretic obstruction each do their own work; neither is a substitute for the other. $\square$

The exact extension theorem leaves room for new mathematics and new physical evidence. It excludes retrospective substitution and lossless recovery from genuinely insufficient input. The next section characterizes the complete evaluated interface itself, rather than assuming that every distinction it carries must already have existed on the proposal side.

5 Qualification, complete observations, and binary reduction

Every actual non-degenerate determinate carrier and evaluated incidence necessarily incurs the kernel. The question here is what information a particular qualification interface must retain. This is settled by the original use and its independently fixed conditions. One Boolean answer can settle one qualification question. Whether that answer also determines every permitted continuation is a further, exact mathematical question.

Fix a set E of evaluated presentations. The set formulation makes all profiles and quotients below ordinary sets; applications using proper classes require the corresponding size conventions. Each presentation has an independently specified original bearer and complete question. A change to a question, a required operation, or an experimental incidence must be identified from its original data, not inferred merely because a proposed answer disagrees with another.

5.1 A binary qualification and its semantic scope

Let $A(e)$ be the original qualification proposition, and write $b : E \rightarrow \{0, 1\}$ for its characteristic value in classical logic. This is a semantic classification. It does not assert an algorithm deciding A , a temporally earlier checking event, or complete access to evidence. A partial reader returning no answer has not thereby returned the negative answer.

A diagnostic for same-side distinctions. Two status values can report the same qualification bit while retaining different further information. The useful diagnostic asks whether their distinction

has a governing effect: it is redescription-dependent when presentations claimed to preserve the same original locus receive discordant required answers; otherwise it is effective if erasing it loses a required answer, and inert if erasing it loses none. A change of raw code alone is not a governing change. Proposition 5.11 makes this diagnostic precise for an adequate realized status interface. It does not infer that every effective distinction is illicit or that every complete interface reduces to one bit.

Proposition 5.1 (Original qualification determines faithful reports). *Suppose two partial readers $r_i : E \rightarrow \{0, 1\}$, $i = 1, 2$, are separately sound for the same original qualification: whenever $r_i(e)$ is defined, it equals $b(e)$. They agree on their common domain. If both have domain E , they equal b extensionally. More generally, equality of their domains and soundness give equality as partial maps.*

Proof. On the common domain both values equal the independently fixed $b(e)$. The domain conclusions then give equality on the entire indicated domain. $\square$

The substantive work is in the original qualification and the individual soundness proofs. Those are not defined by agreement of the readers. An analogous result holds for any independently functional result relation, including non-Boolean outputs. For partial semantic results, an absent complete outcome requires a warrant that no result is incident; failure of a search does not supply that warrant. At a fixed complete locus the actual status has one determined value. Across a family of loci both Boolean possibilities may occur, but neither their joint realization nor uniformity of every further observation follows from the two-element codomain.

Theorem 5.2 (Collective conservation from original result incidences). *Let B be a set of independently identified original data and $\rho : E \rightarrow B$ retain those data. For every required question $q \in Q$, let V_q be its output set and $\mathcal{R}_q \subseteq B \times V_q$ its independently specified, single-valued result relation, possibly partial. Let $h_i : E \rightarrow Z_i$ be a proposed representation and $r_{i,q} : Z_i \rightarrow V_q$ its reader. Suppose each defined reader answer is individually warranted:*

$$r_{i,q}(h_i(e)) = v \implies (\rho(e), v) \in \mathcal{R}_q.$$

Then readers agree whenever they answer the same q on the same original data. If, in addition, each reader covers every actual result incidence,

$$(\rho(e), v) \in \mathcal{R}_q \implies r_{i,q}(h_i(e)) = v,$$

all these representations have the same complete profile of partial answers and definedness, valued in the product of the optional output types $V_q \sqcup \{\perp\}$. Each h_i then determines that profile on its attained image. This conclusion holds simultaneously for an arbitrary set of required questions and output types.

Proof. For any two defined answers concerning equal ρ -data and the same question, soundness places both in the same single-valued relation fibre. Their values therefore agree. For a complete reader, soundness and coverage identify its pulled-back graph exactly with

$$\{(e, v) : (\rho(e), v) \in \mathcal{R}_q\}.$$

This gives both value agreement and exact definedness for every q . If $h_i(e) = h_i(f)$, the same partial reader has the same value and definedness at that shared argument. Thus all original partial-answer coordinates agree at e, f , and the profile is constant on h_i -fibres. The fibre theorem supplies its unique decoding map on $h_i(E)$. Encoding each defined value and its definedness by Boolean characteristic questions gives the equivalent Boolean-profile formulation below. $\square$

This is the positive universal route for complete, accountable redescrptions of the same original object and questions. Pairwise agreement is the conclusion, not a premise. The substantive premises are the original incidence semantics and the individual warrants and coverage of the readers. A purported interface lacking that coverage can still be sound on its answered domain; it cannot thereby claim to preserve the complete interface. Conversely, a complete and individually warranted realization needs no additional permission for its original-incidence conservation. The theorem does not identify different original bearers merely because they share one qualification value, and it asserts semantic determination rather than an effective decoding algorithm.

Corollary 5.3 (Covariant result relations give exact partial equivariance). *Let a group act on B and V_q and suppose*

$$(b, v) \in \mathcal{R}_q \iff (gb, gv) \in \mathcal{R}_q \quad (g \in G).$$

The partial function determined by $\mathcal{R}_q$ has an invariant domain and is equivariant on that domain. Any reader complete and sound for that original relation represents this same partial function on the original data represented by $\rho(E)$.

Proof. An incident result at b transports to one at gb , and applying g^{-1} proves the reverse domain implication. Single-valuedness identifies the transported value as the unique result at gb . The reader conclusion follows from equality of graphs in Theorem 5.2. $\square$

5.2 The canonical interface for a complete use family

Definition 5.4 (Complete observation frame). Let Q be a set of independently specified required Boolean observations $o_q : E \rightarrow \{0, 1\}$. Include an index q_0 with $o_{q_0} = b$. Set

$$P(e) = (o_q(e))_{q \in Q}, \quad e \equiv_P f \iff P(e) = P(f).$$

The family is complete for an intended use family when each required answer and each required application-definedness fact is determined by these observations, with the interpretation justified from the original uses.

Completeness is an application obligation. It is not obtained by deleting observations on which a proposed reduction fails. Boolean coordinates impose no restriction to binary output types: a set-valued output type can be represented by its value-characteristic questions. For partial results, include definedness separately. Observation families may be infinite and need not be effectively enumerable. When a separate original comparison equivalence C must also be respected, include the characteristic observations of its classes; the resulting equivalence is $C \cap \equiv_P$.

Theorem 5.5 (Canonical complete-observation quotient). *The relation $\equiv_P$ is the coarsest equivalence on E preserving every o_q . Its quotient is canonically bijective to $P(E)$. For any interface $h : E \rightarrow Z$, the following are equivalent:*

1. *every o_q factors through h , corestricted to $h(E)$;*
2. *$h(e) = h(f)$ implies $P(e) = P(f)$;*
3. *there exists a unique map $\bar{P} : h(E) \rightarrow P(E)$ with $P = \bar{P} \circ h$.*

If an interface has exactly the fibres of P , its attained image is uniquely isomorphic to $P(E)$ by a map commuting with the original presentation maps.

Proof. Equality of profiles is an equivalence. Any equivalence preserving all coordinates relates only pairs with equal profiles, proving coarseness. The map $[e] \mapsto P(e)$ is well defined, injective, and surjective onto $P(E)$. Coordinate factorizations give equality of profiles on h -fibres. Conversely, assign to $h(e)$ the common value $P(e)$; the fibre condition makes this well defined, and every point of $h(E)$ is attained, proving uniqueness. Coordinate projections give the individual factorizations. If the fibres coincide, the induced map between attained images is also injective and hence bijective; commutation forces it uniquely. $\square$

Theorem 5.6 (Exact criterion for a binary complete interface). *The complete-observation quotient is determined by b alone if and only if*

$$b(e) = b(f) \implies o_q(e) = o_q(f) \quad \text{for every } q \in Q. \quad (*)$$

Under this condition, $\equiv_P = \ker b$ and the quotient is canonically bijective to $b(E) \subseteq \{0, 1\}$. It has exactly two classes if and only if both values occur. Without $()$, two equally qualified presentations have different required observations and cannot be identified by a complete bit-only interface.*

Proof. Condition $(*)$ is precisely the factorization criterion of Theorem 5.5 with $h = b$. Since b is itself a profile coordinate, $P(e) = P(f)$ implies $b(e) = b(f)$. Thus the two kernels coincide exactly under $(*)$. The image and cardinality conclusions follow from the canonical quotient description. Failure of $(*)$ supplies a specific required coordinate separating an equal-bit pair. $\square$

The theorem neither creates a third truth value nor makes kernel governance optional. It distinguishes a single original qualification from a complete family of qualifications and continuation conditions. A domain containing distinct qualified types can require more than their common admission bit. This is not a change of either type by unlawful redescription.

Corollary 5.7 (Binary status with retained base data). *For independently retained data $\rho : E \rightarrow B$, the joint interface $e \mapsto (\rho(e), b(e))$ preserves the complete profile exactly when*

$$\rho(e) = \rho(f), \quad b(e) = b(f) \implies P(e) = P(f).$$

No conclusion identifying presentations with different retained base data follows.

Proof. Apply Theorem 5.5 to the joint map. $\square$

An important realization of this criterion retains an independently complete bearer and the original query interpretations. If the compared outputs are individually warranted against the same functional incidences, conservation supplies their coordinatewise equality. This route establishes preservation collectively without assuming that equal polarity identifies the original object. It also does not make the retained bearer an effectively readable encoding merely by naming it.

5.3 Primitive checks that generate a complete continuation interface

Fix partial operations $t_a : E \rightarrow E$, indexed by a set L , and primitive observations $u_j : E \rightarrow \{0, 1\}$, indexed by J , including b . These operations and observations are specified before any quotient is proposed. For a finite word $w = a_1 \cdots a_n$, let $t_w = t_{a_n} \circ \cdots \circ t_{a_1}$, with t_ϵ the identity. Define Boolean observations

$$d_w(e) = 1 \iff t_w(e) \text{ is defined,}$$

$$v_{w,j}(e) = 1 \iff t_w(e) \text{ is defined and } u_j(t_w(e)) = 1.$$

The pair $(d_w, v_{w,j})$ distinguishes an undefined use from a defined use with negative answer.

Theorem 5.8 (Primitive descent and all finite continuations). *Every observation $d_w, v_{w,j}$ factors through b if and only if both of the following hold:*

1. *every u_j is constant on b -fibres;*
2. *whenever $b(e) = b(f)$, the values $t_a(e), t_a(f)$ are either both undefined or both defined, and in the latter case*

$$b(t_a(e)) = b(t_a(f)) \quad (a \in L).$$

Equivalently, every primitive observation descends and every primitive operation induces a well-defined partial operation on $b(E)$. The resulting induced operations compose to give the induced finite continuations.

Proof. Under the two conditions, define the domain of the induced $\bar{t}_a$ to consist of the attained bit values whose representatives lie in the domain of t_a . Domain saturation makes this independent of the representative. Set $\bar{t}_a(b(e)) = b(t_a(e))$; the output condition proves well-definedness. Define $\bar{u}_j(b(e)) = u_j(e)$ similarly.

Induct on word length. The empty continuation has matching definedness everywhere and preserves the bit. If two equal-bit inputs have matching definedness through w , their intermediate bits agree whenever defined. The primitive conditions then give matching definedness and output bits after the next operation. This proves the induction step. The primitive observation condition gives matching final u_j -values, hence equality of all $d_w, v_{w,j}$ on bit fibres.

Conversely, the empty-word observations give condition (i). For a one-letter word a , equality of d_a gives matching domains. When both values are defined, the observation v_{a,j_0} for $u_{j_0} = b$ gives condition (ii). The definitions of the induced maps and the induction establish composition. $\square$

The theorem constructs the global factorization from primitive domain and result checks. To apply it to all intended standing-bearing uses, the declared finite-context language must cover those uses. Its syntactic induction does not establish that semantic coverage, nor require that arbitrary semantic observations be computable.

Theorem 5.9 (Maximal continuation congruence). *Let $\equiv$ be equality of every $d_w, v_{w,j}$ in the declared finite-continuation profile. Then each primitive t_a has a well-defined partial descent to $E/\equiv$, and every u_j descends. Moreover, $\equiv$ is the coarsest equivalence with these properties: any equivalence preserving the primitive observations, primitive definedness, and primitive output classes is contained in $\equiv$.*

Proof. The empty-word observations preserve every u_j . If $e \equiv f$, the one-letter observation d_a gives matching definedness for t_a . When both outputs exist, each continuation observation at $t_a(e)$ is the corresponding observation for the prefixed word aw at e , and likewise for f . Their equality proves $t_a(e) \equiv t_a(f)$, giving partial descent.

Conversely, suppose an equivalence preserves primitive observations and is a congruence for the partial operations, including their domains. Induction on word length preserves both definedness and the equivalence class of each defined intermediate result. Primitive observation invariance therefore preserves every $d_w, v_{w,j}$. The equivalence is consequently contained in $\equiv$. $\square$

Example 5.10 (A positive split that records a licensed domain). Let $E = \{e_A, e_B, e_R\}$, with $b(e_A) = b(e_B) = 1$ and $b(e_R) = 0$. Let t be the partial identity defined only at e_A . The original type or application condition distinguishes e_A from e_B and is fixed before the interface reduction. The observation d_t separates their common positive bit. The primitive domain condition fails, so the bit-only interface cannot answer every application question. No initial qualification changes, and no representative is promoted by a name. If the original typed objects remain available, the domain distinction can be recovered from them; if only the bit is retained, it cannot.

Proposition 5.11 (Same-side split diagnostic). *Let $\sigma : E \twoheadrightarrow V$ be a realized status interface adequate for the complete profile, so $P = F \circ \sigma$ for a map $F : V \rightarrow P(E)$. Let $\beta(v)$ be the qualification coordinate of $F(v)$, hence $b = \beta \circ \sigma$. For distinct $u, v \in V$ with $\beta(u) = \beta(v)$, let $q : V \rightarrow V/(u \sim v)$ merge just these values. Then P factors through $q \circ \sigma$ if and only if $F(u) = F(v)$, and the decoding map is unique on the attained image.*

Let $\approx$ be the comparison claimed to preserve the original locus and complete questions. Say the split has redescription dependence in governing effect when some $e \approx f$ have status values u, v and $P(e) \neq P(f)$. Exactly one of the following holds:

1. *the split has that redescription dependence;*
2. *it has no such dependence, but $F(u) \neq F(v)$: it is effective, and the merge loses a required profile distinction;*
3. *$F(u) = F(v)$: it is inert, and the merge preserves the complete profile.*

Proof. If $F(u) = F(v)$, define $\overline{F}([z]) = F(z)$. The only non-singleton quotient class is $\{u, v\}$, so the definition is well defined and $P = \overline{F} \circ q \circ \sigma$. Conversely, surjectivity supplies presentations with statuses u and v . A decoding map for their common quotient value must give both $F(u)$ and $F(v)$, forcing equality. Surjectivity onto the attained image also forces uniqueness. Governing redescription dependence implies $F(u) \neq F(v)$. Split first on that dependence, and then, if it is absent, on equality of $F(u)$ and $F(v)$. The alternatives are exhaustive and disjoint. $\square$

If the comparison actually preserves the independently identified original data and the answers have the individual warrants and coverage of Theorem 5.2, governing redescription dependence is excluded. Harmless variation of raw codes with equal decoded profiles belongs to the inert case. An effective split between distinct original inputs can instead retain a warranted type or application-domain difference, as in Example 5.10. Whole-interface bivalence requires all same-polarity values to have equal profiles, precisely the additional factorization condition in Theorem 5.6; the diagnostic alone does not establish it. Retained base data continue to be governed by Corollary 5.7.

6 Standing as the greatest invariant safe interior

Robust eligibility for a specified repertoire asks a different complete question from immediate entry. The repertoire and the original warrant obligations must therefore be fixed before robust standing is classified. Calling only the ultimately successful paths “licensed” and then quantifying over those paths would omit the very failures the classification is meant to detect.

Definition 6.1 (Independent continuation data). Let X be a set of configurations, L a set of operation labels, and $R \subseteq X \times L \times X$ a declared continuation relation. Give an independent local

qualification predicate $A \subseteq X$ and an edge-warrant predicate $W(x, a, y)$ on R . A finite path is a sequence

$$x_0 \xrightarrow{a_1} x_1 \xrightarrow{a_2} \dots \xrightarrow{a_n} x_n$$

whose triples lie in R . It is safe when every x_i belongs to A and every edge satisfies W .

Here R lists the applications whose safety the claimed repertoire undertakes to sustain. It may be nondeterministic or infinitely branching. Its edges are not defined by the standing predicate constructed below. The predicates A and W record actual target, typing, domain, protocol, or proof conditions, with their independent semantic interpretations. Calling them warrants does not prove their soundness.

Optional guarded operations need not be available at every state. If operation labels in a specified set $M(x) \subseteq L$ are required at x , include the readiness condition

$$(\forall a \in M(x))(\exists y \in X) (x, a, y) \in R$$

in $A(x)$ alongside the original local qualification. This prevents a missing required operation from passing merely because no outgoing edge exists. Safety by itself imposes neither termination nor a fairness or progress condition; such a requirement needs its own independently stated semantics. The theory applies equally to physical protocols and atemporal proof or construction dependencies.

Definition 6.2 (Safe interior and its operator). Let $S \subseteq X$ contain exactly those x from which every finite declared path is safe, including the path of length zero. For $U \subseteq X$ define

$$\Phi(U) = \{x \in A : (\forall a, y)((x, a, y) \in R \implies W(x, a, y) \text{ and } y \in U)\}.$$

A sound inductive candidate is a subset U with $U \subseteq \Phi(U)$.

Theorem 6.3 (Greatest safe interior). *The operator Φ is monotone and*

$$S = \Phi(S) = \bigcup \{U \subseteq X : U \subseteq \Phi(U)\}.$$

Consequently S is the greatest fixed point and the greatest sound inductive candidate. It exists without finiteness, finite branching, a choice of paths, or an effective presentation.

Proof. If $U \subseteq V$, membership in $\Phi(U)$ implies membership in $\Phi(V)$, proving monotonicity. If $x \in S$, the zero-length path gives $x \in A$. Every outgoing edge has its warrant, since it is a one-edge path from x . Every finite path from its successor can be prefixed by that edge; its safety follows from $x \in S$. Thus each successor lies in S , and $x \in \Phi(S)$.

Conversely, let $x \in \Phi(S)$. The zero-length path is safe because $x \in A$. A nonempty path starts with an edge having its warrant and reaching a point of S ; its suffix is safe by the definition of S . Thus $x \in S$.

For a sound inductive U , induction along any finite path from $x \in U$ keeps each state in $U \subseteq A$ and supplies every edge warrant. Hence $U \subseteq S$. Since S is itself a sound inductive candidate, it equals the union of all of them. Every fixed point is a sound inductive candidate, so S is the greatest fixed point as well. The argument quantifies over given finite paths and chooses no infinite branch. $\square$

Corollary 6.4 (Positive realizability and invariant witnesses). *For every $x \in X$, membership $x \in S$ is equivalent to the existence of a sound inductive candidate U containing x . In particular, $S \neq \emptyset$ exactly when there exists a nonempty $U \subseteq A$ whose every declared outgoing edge has its warrant and returns to U .*

Proof. If $x \in S$, take $U = S$. Conversely every sound inductive candidate lies in S by Theorem 6.3. The nonempty formulation spells out $U \subseteq \Phi(U)$. $\square$

Existence of the greatest set does not by itself establish that the requested robust repertoire is realizable. The interior can be empty even when immediate qualification has positive instances: one qualified state whose required continuation reaches an unqualified state is enough. A nonempty independently verified invariant region supplies the missing positive witness. An actual determinate incidence is still kernel-governed when it fails a proposed robust-use requirement.

Corollary 6.5 (The precise uniqueness and completeness claim). *Every sound inductive candidate is contained in S . A sound inductive candidate is complete for robust qualification exactly when it equals S . Every inclusion-maximal sound inductive candidate equals S , so there is exactly one such maximal candidate. No uniqueness of arbitrary sound proper subsets is asserted.*

Proof. Containment follows from Theorem 6.3. Completeness adds the reverse containment. Because S is itself a sound inductive candidate containing every other one, any inclusion-maximal candidate must equal it. $\square$

An arbitrary claimed complete interior I can fail by surplus, $I \setminus S \neq \emptyset$, or by omission, $S \setminus I \neq \emptyset$, or by both. Absence of surplus alone does not exclude omissions. For example, with two qualified states and identity transitions, each singleton is sound but incomplete; their union is the unique greatest interior.

Theorem 6.6 (Finite-depth approximation and finite failure witnesses). *Put $U_0 = X$ and $U_{n+1} = \Phi(U_n)$. Then*

$$U_0 \supseteq U_1 \supseteq \cdots, \quad S = \bigcap_{n \geq 0} U_n.$$

No branching restriction is required. If $x \notin S$, some finite declared path from x has a locally unqualified state or an unwarranted edge. If X is finite, the descending sequence stabilizes after at most $|X|$ strict removals and its stable value is S .

Proof. The first inclusion is automatic and monotonicity gives all subsequent inclusions. For any nonempty family (V_i) ,

$$\Phi\left(\bigcap_i V_i\right) = \bigcap_i \Phi(V_i) :$$

the local and edge-warrant requirements are unchanged, and saying that every successor belongs to every V_i is equivalent to saying it belongs to their intersection. Therefore, for $U_\infty = \bigcap_{n \geq 0} U_n$,

$$\Phi(U_\infty) = \bigcap_{n \geq 0} U_{n+1} = U_\infty.$$

The greatest-fixed-point theorem gives $U_\infty \subseteq S$. The reverse inclusion follows by induction from $S = \Phi(S) \subseteq U_n$ for every n .

For the failure witness, the definition of S says that all finite paths are safe. Its negation in classical logic gives a finite path failing either a node or an edge condition. Equivalently, if $x \notin U_{n+1}$, then either $x \notin A$, an outgoing edge lacks its warrant, or some successor is outside U_n ; induction yields a finite witness in each case. No infinite branch is required. On a finite state set there can be at most $|X|$ strict decreases. Once equality occurs, applying Φ repeats the same set, which is a fixed point containing S and hence equals S . $\square$

For a finite, explicitly supplied graph and decidable local predicates, the descending construction is an algorithm. Arbitrary sets and semantic predicates need not give an effective search procedure. The existence of a finite counterexample when safety fails does not assert that an unrestricted program can find it, or that all successful safety claims have finite certificates.

Infinite paths and local safety. The relation R already permits arbitrary branching. For this universal-successor operator, the intersection identity in the proof of Theorem 6.6 gives the stated countable descending approximation without a finite-branching hypothesis. Every infinite path from $x \in S$ satisfies all local qualification and edge-warrant conditions: each particular node or edge occurs in a finite prefix, which is safe by definition. This does not make testing only infinite paths sufficient. A qualified state with one warranted edge to an unqualified deadlock has an unsafe finite path and no infinite path; the infinite-path-only test is vacuous. The finite-path definition therefore retains failures at terminal states as well as failures along indefinitely continuing paths.

Theorem 6.7 (When stable standing equals immediate admission). *Let s be the characteristic predicate of S . Then $s(x) = 1$ implies $x \in A$. Moreover,*

$$S = A \iff (\forall x \in A)(\forall a, y)((x, a, y) \in R \implies W(x, a, y) \text{ and } y \in A).$$

Thus identification of stable standing with the initial qualification is exactly a local preservation and edge-warrant theorem for the declared repertoire.

Proof. The zero-length path gives $S \subseteq A$. The right-hand condition says $A \subseteq \Phi(A)$, so greatestness gives $A \subseteq S$. Conversely, if $A = S$, the fixed-point equation $S = \Phi(S)$ gives every displayed successor and warrant condition. $\square$

The standing role is not created by this theorem. The theorem characterizes the actual qualification for the specified robust-reuse question and tells a purported implementation what to establish. A complete and sound implementation of that question agrees with s . An initial entry question has the same extension only when the displayed preservation condition holds.

6.1 Invariance and faithful transport of the interior

Theorem 6.8 (Lawful redescription and relation-isomorphism transport). *Consider two independent continuation systems (X, L, R, A, W) and (X', L', R', A', W') . Suppose bijections $h : X \rightarrow X'$ and $k : L \rightarrow L'$ satisfy*

$$x \in A \iff h(x) \in A', \quad (x, a, y) \in R \iff (h(x), k(a), h(y)) \in R',$$

and corresponding edges have the same warrant truth value. Then $h(S) = S'$. More generally, within one system an equivalence $\sim$ preserves S whenever it preserves A and matches outgoing transitions in both directions, with equivalent successors and the same edge-warrant values.

Proof. For the isomorphism, applying h and k to a finite path gives a finite path in the other system and preserves every safety condition. The inverse bijections give every path back, so safety from x is equivalent to safety from $h(x)$.

For the equivalence formulation, suppose $x \sim x'$ and a finite path from x' fails safety. Match its first transition at x and continue by finite induction, using the equivalence of successors at each step.

Local qualification and edge-warrant truth values agree on the matched path, so the resulting path from x also fails. Thus $x \in S$ implies $x' \in S$; symmetry gives the converse. Finite path matching requires no global choice of paths. $\square$

Source-to-target qualification transfer can also be proved without an isomorphism. A useful sufficient condition is that h preserves local qualification and every target transition from $h(x)$ lifts to a source transition from x , with mapped successor and preserved warranted edges. Then $h(S)$ is a sound inductive target candidate and lies in S' . Coverage of target transitions matters: mapping old safe edges into a richer target does not exclude an additional unsafe target edge. A proved transfer supplies a target warrant; merely copying a positive source bit does not.

7 Conservation, new constructions, and retained history

Three conservation statements have different subjects. At an unchanged original bearer and complete question, every individually warranted answer has the same value. Along a declared continuation from the safe interior, robust positivity is preserved. Along a retained history, the original qualifications of earlier incidences remain what they were. These statements require no identification of a new output with its input.

Proposition 7.1 (Safe continuation and sufficient certificates). *If $x \in S$ and $(x, a, y) \in R$, then $W(x, a, y)$ and $y \in S$. Any independently justified predicate $C(x)$ with $C(x) \Rightarrow x \in S$ is a sound sufficient certificate for robust qualification; it need not cover every point of S . A complete replacement classifier additionally requires $x \in S \Rightarrow C(x)$.*

Proof. The edge conclusion is the fixed-point equation. The certificate assertions are respectively soundness and soundness together with completeness for the original robust-qualification proposition. $\square$

A verified constructor may establish a new qualified output from its typed premises. Its target warrant can be supplied by the constructor theorem rather than by a second checking event. This is compatible with the absence of an unsupported source of standing. Likewise, a correct report may replace an incorrect report without changing the original incidence; both discordant reports cannot be individually warranted for that incidence.

Theorem 7.2 (Retained history and the exact endpoint condition). *Let $c : X \rightarrow \{0, 1\}$ be an independently fixed qualification, and define the all-stages qualification of a finite path by*

$$H(x_0, \dots, x_n) = 1 \iff c(x_i) = 1 \text{ for every } 0 \leq i \leq n.$$

Extending a retained path preserves every original $c(x_i)$, and H cannot change from 0 to 1. The following are equivalent:

1. *for every finite R -path, $c(x_n) = 1$ implies $H(x_0, \dots, x_n) = 1$;*
2. *negative qualification is absorbing: every $(x, a, y) \in R$ satisfies $c(x) = 0 \Rightarrow c(y) = 0$.*

Proof. A path extension retains its original entries and their independently fixed qualifications. A failed entry therefore remains a witness that $H = 0$. If (i) holds, apply it to a one-edge path: a positive endpoint cannot have a negative predecessor, giving (ii). Conversely, under (ii), any

negative entry propagates forward along a finite path by induction, so a positive endpoint excludes all negative entries. This gives (i). $\square$

The theorem is exact for the stated transitions. Kernel governance and historical retention do not supply negative absorption for an arbitrary state predicate. For example, let $X = \{r, s\}$, $A = \{s\}$, and R contain $r \rightarrow s$ and $s \rightarrow s$, with true edge warrants. Then $S = \{s\}$ and a nonstanding state has a standing successor. The history beginning at r retains its failed all-stages qualification. No original negative answer has become positive.

If a stronger conservation claim requires positivity and negativity both to be preserved across the same transitions, it must establish both directions. If it instead concerns one unchanged complete incidence, exact value preservation follows directly from original-incidence accountability. The subject and transition conditions determine which conservation statement applies.

8 Semantic exhaustion and impossibility of same-scope evasion

An exhaustion theorem must cover the intended failures before assigning them names. Fix the original bearer, complete questions, required uses, and their semantic conditions independently of the proposed implementation. We first prove exact alternatives for answers, information, and robust reuse. We then locate five failure mechanisms within those alternatives. The resulting impossibility is substantial: an incorrect original answer cannot become correct by faithful recoding, missing information cannot be recovered by postprocessing, and an unsafe claimed interior cannot evade its failed original condition. None of this makes kernel governance optional for a failed construction or its determinate carrier.

8.1 Original result graphs: surplus, omission, and negative claims

Use the data of Theorem 5.2. For each original required question $q \in Q$, pull its independently specified result relation back to the evaluated presentations:

$$G_q = \{(e, v) \in E \times V_q : (\rho(e), v) \in \mathcal{R}_q\}.$$

Thus G_q is single-valued, possibly partial. Let $K_q \subseteq E \times V_q$ be the graph of answers a proposal asserts, after interpreting its outputs as answers to that original question. We initially allow inconsistent proposals, so K_q need not be single-valued. Soundness requires $K_q \subseteq G_q$. If the proposal undertakes complete coverage on $M_q \subseteq E$, it additionally requires

$$G_q \cap (M_q \times V_q) \subseteq K_q.$$

Full original coverage has $M_q = E$. Omitting a result outside the advertised coverage is not thereby a violation. Silently replacing an original requirement for full coverage by a smaller M_q does not discharge that requirement.

An assertion of semantic absence needs a separate interpretation. Write $\perp_q$ for an extra symbol and set

$$G_q^\perp = G_q \cup \{(e, \perp_q) : e \notin \text{dom}(G_q)\}.$$

This is the graph of a total semantic outcome in $V_q \sqcup \{\perp_q\}$. A claimed absence contributes the record $(e, \perp_q)$; a computation that has not answered contributes no such record. A Boolean negative

answer $0 \in V_q$ is an ordinary result, different from $\perp_q$. The totalization describes complete outcome semantics; it does not turn an arbitrary computational nonanswer into a correct absence report.

Theorem 8.1 (Exact original-graph exhaustion). *For every proposed complete original result graph K_q , exactly one of the following set-theoretic alternatives holds:*

1. $K_q = G_q$;
2. $K_q \setminus G_q \neq \emptyset$ and $G_q \setminus K_q = \emptyset$;
3. $G_q \setminus K_q \neq \emptyset$ and $K_q \setminus G_q = \emptyset$;
4. both differences are nonempty.

A surplus record is an incorrect original answer. An omitted record violates claimed complete coverage. The same alternatives hold for explicit complete-outcome graphs compared with $G_q^\perp$. There a surplus absence record asserts nonexistence of a result that actually occurs, while an omitted absence record violates a claim to answer every complete-outcome question. These statements hold simultaneously for arbitrary set-sized families Q ; a failure of their simultaneous equality has a witness at some original question and presentation.

Proof. Two sets are equal exactly when both directional differences are empty. The four alternatives enumerate the two independent emptiness propositions and are mutually exclusive. If $(e, v) \in K_q \setminus G_q$, the original incidence does not warrant that answer. If $(e, v) \in G_q \setminus K_q$, an actual result is missing from the advertised complete graph. For totalized outcomes, $(e, \perp_q) \notin G_q^\perp$ means exactly $e \in \text{dom}(G_q)$, which gives the stated absence error. Quantifier negation gives a coordinate q when equality fails for the whole family, and the corresponding nonempty difference supplies its record. For restricted coverage, restrict the omission test to $M_q \times V_q$; the soundness test still concerns every asserted record. $\square$

The theorem catches an invariant but initially wrong verdict: a classifier returning 1 everywhere has a surplus wherever the original qualification is 0, even though no relabeling changes its answer. Conversely, a classifier returning 0 at an actually qualified incidence has an incorrect negative result. No history of prior rejection is needed. A sound partial checker that advertises no completeness can omit positive instances without making either mistake. A declaration that the same checker covers all original instances adds a distinct, testable claim.

For the bridge, “supported the specified load without auxiliary support” fixes the question before the trial is classified. A false positive and a false negative are both detected by original incidence. Failure to locate a photograph or certificate does not itself show that the trial failed. A later warranted measurement can establish a previously unknown fact without changing that fact.

8.2 Information and interiors: exact alternatives with witnesses

Let $P : E \rightarrow \prod_{q \in Q} O_q$ be the complete original outcome profile, using $O_q = V_q \sqcup \{\perp_q\}$ when definedness is required. Its coordinates are fixed by the original questions, not selected to fit a proposed reduct $h : E \rightarrow Z$.

Proposition 8.2 (Complete-interface alternative). *Exactly one of the following holds:*

1. there is a unique decoder $\overline{P} : h(E) \rightarrow P(E)$ with $P = \overline{P} \circ h$;

2. there exist $e, f \in E$ and $q \in Q$ such that $h(e) = h(f)$ but $P(e)_q \neq P(f)_q$.

In the second case no deterministic procedure receiving only $h(e)$ can return all original outcomes correctly on all presentations. In the first case a proposed decoder can still be wrong, incomplete, or unsupported; existence of the semantic decoder does not certify that implementation.

Proof. Theorem 4.1 applied to P makes (i) equivalent to profile constancy on every h -fibre. Negating constancy gives an equal-input pair with different profiles; equality in a product is coordinatewise, so some q distinguishes them. A correct decoder at their shared input would have to give both different outcomes, which is impossible. Conversely, such a pair contradicts any factorization. Uniqueness is on the attained image; neither computation nor values outside that image are asserted. $\square$

An omitted *syntactic* observation is not necessarily lost information: it may be determined by retained observations. To check a proposed reduced family $Q_0 \subseteq Q$, apply this proposition to $h = P|_{Q_0}$. A genuinely missed requirement has a required coordinate not determined by the reduced profile, with the displayed witness. A proof concerning only a declared subfamily does not establish coverage of all original uses. That further assertion requires an interpretation of each original use in the retained family, with preservation of its result and definedness. If that interpretation has not been supplied, completeness is unestablished; its failure has not yet been proved.

Next retain the independently fixed X, L, R, A, W of Definition 6.1 and its greatest safe interior S . Let $I \subseteq X$ be any claimed complete robust interior.

Theorem 8.3 (Exact interior exhaustion and local failure witnesses). *Either $I = S$, or I has surplus $I \setminus S \neq \emptyset$, or omission $S \setminus I \neq \emptyset$, or both. Every surplus point starts a finite declared path with an unqualified node or an unwarranted edge. Every omitted point belongs to a sound inductive candidate not contained in I . Conversely, such an omitted point witnesses incompleteness, and such an unsafe path from a point of I witnesses unsoundness. Hence $I = S$ exactly when*

$$I \subseteq \Phi(I) \quad \text{and} \quad (\forall U \subseteq X)(U \subseteq \Phi(U) \implies U \subseteq I).$$

Proof. The set differences give the alternatives. If $x \in I \setminus S$, Theorem 6.6 supplies a finite unsafe path. If $x \in S \setminus I$, the sound candidate $U = S$ contains x ; conversely, a point of any sound inductive U lies in S by greatestness. An unsafe path contradicts membership in S directly. For the final equivalence, the first inclusion gives $I \subseteq S$; the second, applied to $U = S$, gives $S \subseteq I$. Equality implies both by the fixed-point equation and greatestness. $\square$

One may select the earliest failed node or edge on a supplied finite path by ordering its finite sequence of node and edge tests. This gives a primary diagnostic for that path, not disjoint underlying causes. A qualified bridge whose required continuation removes essential support can witness surplus. An excluded, independently certified safe design can witness omission. Absence of unsafe examples proves neither completeness nor nonempty realizability. Required readiness must already be included in the original local condition as in Definition 6.1; otherwise a missing operation is not detected by checking only existing edges. Eventual success and fairness need their own original semantics and do not acquire finite failure witnesses from this safety theorem.

8.3 Coverage before the five mechanism names

The scope of exhaustion can now be stated without a registry of implementation styles. An *original requirement frame* contains: the independently interpreted bearer and question identities; the

required result relations G_q ; the continuation data R, A, W ; and, where an argument is offered as a certificate, the interpreted premises and rule conditions of Definition 4.3. It is specified before evaluating the proposal. Its questions include any required reference comparisons, entry qualifications, operation-definedness facts, and original historical claims. It does not include a new predicate defined merely as “this proposal passes the five tests.”

Lemma 8.4 (Constructive coverage of a specified original target). *Let U be any independently specified set of required original questions. On their evaluated presentations E , let each $u \in U$ have an independently interpreted single-valued relation $G_u \subseteq E \times V_u$, possibly partial. There exists a complete semantic profile for this entire target family, without a finite list of question forms or a computability assumption. Every original partial result graph is recovered from its coordinate. The same profile can be represented by a set of Boolean observations. Any complete faithful realization of these graphs through an interface h determines this profile uniquely on $h(E)$.*

Proof. The totalized relation $G_u^\perp$ defines a unique outcome $a_u(e) \in V_u \sqcup \{\perp_u\}$ for each e . This uses the original relation, not the proposal’s answer. Form

$$P_U(e) = (a_u(e))_{u \in U}.$$

The coordinate at the original index u is exactly its semantic outcome; its non- $\perp_u$ entries recover G_u . Thus identity indexing of U proves coverage, without selecting a favorable subfamily. Unique outcomes define the function without selecting among several possible results. For Boolean coordinates take all pairs (u, v) with $v \in V_u \sqcup \{\perp_u\}$ and the truth values of $a_u(e) = v$. They determine each outcome and domain. Complete, individually sound partial readers through h have exactly the graphs G_u by Theorem 5.2, so their values and definedness determine P_U . It is consequently constant on h -fibres, and Theorem 4.1 gives its unique decoder on the attained image. $\square$

Totalization here represents a partial graph mathematically; it requires an explicit absence report only where the original question or a completeness undertaking requires that report. The original coverage domains M_q remain in force. A checker undertaking soundness alone is not required to realize the complete profile. Typed input carriers can be tagged in a disjoint union, with each question restricted to its intended inputs. Required local qualification, edge warrants, readiness, reference comparisons, and retained-history answers enter as independently interpreted original questions. A required certificate-acceptance condition also needs its own interpretation; it is not inferred from the truth of the conclusion. The lemma supplies a canonical complete semantic frame for every specified target, not an effective oracle or a proof that a particular inventory exhausts an otherwise unspecified physical problem. A relational specification with several permitted outputs can be represented by its original Boolean membership questions; no arbitrary choice of one permitted output is required.

There are two distinct completeness duties. *Requirement coverage* means that these original questions and uses cover the intended problem, with an independently justified interpretation. *Realization coverage* means that the proposal answers or realizes all the requirements it undertakes. The first cannot be proved by the second. For a generated use language, an interpretation proof proceeds from the specified primitive operations and observations through its formation rules; Theorem 5.8 supplies the finite-continuation induction once those primitives and domains are covered. An application containing additional physical observations, progress requirements, or inference principles must include their interpretations too. Calling the vocabulary exhaustive does not supply this proof.

Lemma 8.5 (Localization of a false certificate conclusion). *Consider a well-founded interpreted proof tree for an original requirement. If its conclusion is false under the original semantics, some leaf is false or some inference instance has true premises and a false conclusion. If adjacent uses of a premise change its interpreted bearer or question, that change instead requires a justified translation; it cannot be treated as the same premise by its label alone.*

Proof. Among nodes with false interpreted conclusions, choose a minimal one in the well-founded dependency order. If it is a leaf, the first alternative holds. Otherwise its immediate premises are true by minimality while its conclusion is false, giving the second. This instance cannot satisfy the claimed semantic implication. The argument presupposes that each dependency conveys its actual interpreted premise; substituting another interpretation changes that premise and is not an instance of the inference being checked. A valid translation can be included as an independently warranted inference. $\square$

This is a localization of semantic error, not a claim that every actual warrant comes with a supplied proof tree. A true conclusion may have a defective offered proof or no proof presently known. A cyclic certificate can be accepted through a separately justified induction or coinduction principle; the cycle alone does not establish that principle. Theorem 4.8 additionally excludes authorization by a rule closure with no grounded entrance. It does not exclude an independent entrance that the proposed dependency sketch omitted.

These tests define the following five failure mechanisms. They may overlap, and their number is not asserted to be minimal.

F1: Silent redescription.

A claimed identity-preserving change alters a required original answer or its asserted applicability without the required interpretation. For faithful interpretations of the same functional original incidence, discordant answers force a surplus answer or a coverage defect. Different amounts of evidence alone need not be a semantic change.

F2: Incorrect qualification or unfulfilled admission obligation.

An original result or use record is attributed a status contrary to its independently fixed incidence; an unqualified use is admitted; or a claim of complete coverage omits a required original incidence. Qualification of an answer record (e, q, v) here means $(e, v) \in G_q$, fixed before the proposal. This includes false negatives and initially wrong invariant verdicts. A faulty premise or inference offered to warrant entry is tested against its original semantic conditions. An unsupplied warrant or coverage proof is recorded as unestablished, not as proof of negative physical qualification. No temporally separate inspection is required.

F3: Retrospective substitution.

A later result is used to alter the attributed qualification of the same original occurrence or its retained history. A corrected report of that occurrence can be lawful; substituting a repaired occurrence for it changes the subject. Theorem 7.2 detects the retained-history error without prohibiting a qualified new endpoint.

F4: Unwarranted scope transport.

An answer or certificate for another bearer, question, or repertoire is applied to the original one without a valid interpretation or transfer. A proved implication can provide a sufficient certificate, and Theorem 6.8 supplies a full transport criterion. Coinciding labels, endpoints, or positive bits supply neither result on their own.

F5: Unlicensed distinguishing information.

A claimed answer from an advertised lower interface depends on a distinction that interface does not retain, or a purported content-determined selection violates its independently established symmetry. Proposition 8.2 and Theorem C.6 give the corresponding obstructions. An independently justified added channel or anchor can change the available problem. Metrics, probabilities, and auxiliary parameters are not failures merely by their names; the defect is their asserted authority for an original claim without the required interpretation or information.

Theorem 8.6 (Failure-class exhaustion for the original use problem). *Fix an independently specified original target family and its required-use and dependency semantics. Lemma 8.4 constructs its complete semantic requirement frame; a reduced presentation must establish the corresponding coverage. Interpret a proposal's original answer records, claimed coverage, proposed complete interior, and offered certificate dependencies in this frame. Then every semantic violation of those original requirements has one of the following proved witnesses:*

1. *an incorrect original-locus attribution or an invalid transport between interpreted loci;*
2. *a surplus answer or an omission from claimed original coverage;*
3. *a surplus interior point with a finite failed node or edge, or an omitted point belonging to a sound inductive region;*
4. *a false premise or an inference instance not preserving its true original premises, when a false conclusion is offered through a well-founded certificate.*

These witnesses instantiate F1–F5 as specified above, with F2 covering original qualification and coverage errors independently of their cause. A semantic decoder on the attained lower-interface image further exists exactly under the fibre condition; its failure has the distinguishing-pair witness. No additional implementation style can evade these alternatives while preserving the covered original semantics. This is a coverage theorem for the stated requirement frame, not a claim that five names establish coverage of an unspecified domain.

Proof. First compare the interpreted bearer, question, and occurrence with the independently required original locus. An unauthorized change is (i); an accepted translation instead places the resulting claim back at its required original locus. Every original question has its own coordinate by Lemma 8.4; a reduced frame supplies that coordinate through its coverage proof. Thus each answer claim has a definite q and an interpreted output, and Theorem 8.1 covers correctness and the advertised coverage. No implementation syntax enters this argument. The robust-safety part has the fixed path semantics of R, A, W ; Theorem 8.3 gives (iii), reducing surplus to original node or edge conditions and omission to a missed sound region. Other original requirements, including any liveness or eventual-outcome question, remain graph coordinates under (ii); they do not receive a finite-path witness from (iii). Where correctness is claimed through a well-founded certificate, Lemma 8.5 gives (iv). Coverage is supplied by the original-index construction or a proved reduction of it, not inferred from the mechanism names.

The witness in (ii), or a failed original node or edge in (iii), is F2 by the independently specified incidence or use predicate. Omission is F2 only where completeness was claimed or required. A false premise or unsound inference offered as warrant fails the corresponding F2 support obligation. An original-locus substitution in (i) is F1 when attributed to redescription, F3 when a later occurrence replaces the retained original, and F4 when another question or scope supplies the asserted authority. These descriptions can coincide. If the proposal instead promises recovery from a lower interface,

Proposition 8.2 proves the decoder alternative; using an unretained discriminator as if it belonged to that interface is F5. A wrong fixed choice still gives the direct F2 graph witness. Thus the mechanism vocabulary is anchored in the preceding witnesses, rather than used to prove their existence. $\square$

The theorem makes the residual obligation explicit. If the intended use family has not been related to the original frame, the advertised universal exhaustion has not been established. This does not supply a counterexample to the covered theorem, prove that the proposal is physically unqualified, or license a missing observation to be deleted. One must establish that relation or state the still-uncovered requirement. Likewise, if all semantic conditions actually hold but their proofs are not yet supplied, the implementation may be correct while its offered justification remains incomplete.

8.4 Impossibility dispositions and the local gate

Corollary 8.7 (Same-scope fail-closed consequence). *A witnessed surplus answer cannot be retained as a correct answer to the same original question. A witnessed omission cannot be retained together with the corresponding claim of complete coverage. An unsafe surplus cannot remain in a sound complete interior, and an omitted safe point cannot remain excluded from an interior claimed complete. A discordant fibre excludes every universally correct decoder using only that lower input. If $S = \emptyset$, no constructor can realize a positively qualified robust output for this unchanged repertoire. A genuinely unseeded authorization closure cannot establish the omitted target authority.*

Proof. The first two claims contradict the respective set inclusions of Theorem 8.1. The next two contradict equality with S by Theorem 8.3. The decoder claim is Proposition 8.2, including the impossibility for all postprocessing from Corollary 4.2. An output in an empty S is impossible; Corollary 6.4 gives the exact positive-witness criterion when realizability is claimed. The authorization claim is Theorem 4.8. None is removed by renaming the witnessed record while retaining its original interpretation. $\square$

The dispositions follow the witness. Withdraw or correct the false report; supply a valid missing result or restrict an unjustified completeness claim; remove unsafe proposed uses and establish the revised repertoire; include omitted safe cases when completeness is claimed; or add independently licensed information and acknowledge the resulting change of input. A changed target is a new stated problem, not a solution of the unchanged one. With no conclusive witness but an undischarged proof obligation, withhold the unsupported assertion of certification. These conclusions do not require rejecting the underlying physical object, nor equating an unresolved computation with a negative result. A correct report that the requested positive realization is impossible remains a correct answer; it is not itself a realization of that positive request.

The prior local gate is consequently a requirement of original-incidence accountability, not a mandatory earlier checking event. A use asserted to bear standing must actually satisfy its original qualification; an offered certificate must warrant that application. A constructor theorem can supply the warrant within the construction itself. Deferring human discovery does not defer the occurrence's actual qualification, and later success does not retrospectively authorize a different failed occurrence. Proposition 2.3 supplies the necessary functional roles; the exhaustion theorems show exactly why these represented original obligations admit no sixth semantic escape once coverage is proved. They neither derive that coverage from five labels nor forbid new faithful constructions.

9 Worked realizations and discriminating examples

The examples exhibit both successful realizations and concrete failures of the mathematical tests. Each challenged claim is paired with an original result, a lost observation, or a continuation witness that settles it. This connects the examples directly to the graph, interface, and interior alternatives of Section 8.

9.1 A finite bridge protocol

Consider an idealized four-state protocol. It is a mathematical model of the distinctions in Section 1.1, rather than an empirical engineering law. States describe current configurations, not immutable names for a single original trial:

- s : an intact configuration that supports the stated current load and remains qualified after the specified repeat;
- t : a configuration that supports the current load but fails under the specified repeat;
- f : a failed configuration, to which the protocol can apply a repair;
- r : the repaired configuration after a new successful trial, stable under the specified repeat.

Let $A = \{s, t, r\}$ and declare the transitions

$$s \xrightarrow{\text{repeat}} s, \quad t \xrightarrow{\text{repeat}} f, \quad f \xrightarrow{\text{repair}} r, \quad r \xrightarrow{\text{repeat}} r.$$

All four edge warrants are true: the named operations concern the correct configurations and prescribed conditions. This does not assert that every operation preserves current load qualification. That preservation is a separate question for A .

Configuration	Current qualification	Next configuration	U_1	S
s	1	s	yes	yes
t	1	f	yes	no
f	0	r	no	no
r	1	r	yes	yes

The finite-depth construction gives

$$U_0 = \{s, t, f, r\}, \quad U_1 = A, \quad U_2 = \{s, r\} = U_3 = S.$$

Write $b(x) = 1$ exactly when $x \in A$ and $b_{\text{reuse}}(x) = 1$ exactly when $x \in S$. These answer different original questions: current load qualification and qualification throughout the declared reuse protocol. Let

$$G_{\text{reuse}} = \{(x, b_{\text{reuse}}(x)) : x \in X\}$$

be the actual robust-qualification graph. The following proposed claims can now be tested explicitly.

Claims, witnesses, and failed tests.

1. *A surplus record and an unsafe admission.* Suppose a proposed reader asserts $(t, 1)$ as a robust-qualification record. The actual graph contains $(t, 0)$, so

$$(t, 1) \in K_{\text{reuse}} \setminus G_{\text{reuse}}.$$

This is a surplus answer under Theorem 8.1. Its concrete reason is the path $t \xrightarrow{\text{repeat}} f$, whose endpoint fails local qualification. Equivalently, a proposed complete interior $I_+ = \{s, t, r\}$ contains the unsafe surplus $t \in I_+ \setminus S$, witnessed by that same path under Theorem 8.3. The failed assertion concerns robust reuse; the current-load answer $b(t) = 1$ remains correct. The failure at f itself has a zero-length witness.

2. *An omitted safe design and an incomplete interior.* Suppose $I_- = \{s\}$ is presented as the complete interior. It is qualified and closed under its warranted self-loop, so it passes the soundness test. Nevertheless $r \in S \setminus I_-$: the repaired configuration also qualifies and its declared repeat returns to r . Thus $\{r\}$ is an independently inspectable sound invariant region omitted by I_- . This is the completeness failure of Theorem 8.3, with the omitted safe design as its witness. If a total classifier reports 0 at r because $r \notin I_-$, it additionally asserts the incorrect record $(r, 0)$ and omits the correct record $(r, 1)$. A partial certificate reporting only $(s, 1)$ remains sound; its omission is a defect only if it claims or is required to provide complete coverage.
3. *A fibre collision.* On the domain of the repeat operation, let o_{repeat} ask whether the resulting configuration qualifies. Then

$$b(s) = b(t) = 1, \quad o_{\text{repeat}}(s) = 1, \quad o_{\text{repeat}}(t) = 0.$$

Any decoder receiving only the current qualification bit has the same input at s and t but would need two different outputs. This pair defeats the fibre criterion of Proposition 8.2 and the binary-reduction criterion of Theorem 5.6. It also shows why the effective same-side distinction in Proposition 5.11 must be retained for this use family. An adequately interpreted configuration or its complete profile retains the distinction.

The positive construction is equally explicit. Both $\{s\}$ and $\{r\}$ are sound invariant candidates, and their union $S = \{s, r\}$ is the complete interior. The descending calculation above includes every safe configuration and supplies witnesses excluding t and f . Thus the example tests exclusion and coverage together, as required by Theorem 6.3 and Corollary 6.5.

The edge $f \rightarrow r$ disproves negative absorption for current qualification. It does not repair the original failed trial. A history retaining f still contains that failure after the repaired trial occurs, so its all-stages predicate remains zero. Each trial and repair is determinate and kernel-governed. The model thereby realizes stable standing, its unique complete interior, and historical conservation while separating them from a false ban on the construction of a new qualified state.

9.2 The finite data actually needed for a successful bit interface

The negative example has a positive counterpart. Suppose X is partitioned into nonempty sets X_+ and X_- , the qualification bit marks that partition, every primitive observation is a specified function of the bit, and every declared partial operation has a domain that is a union of bit classes and sends each domain class into a single bit class. These statements can be verified directly from a

transition table. Theorem 5.8 then proves descent of all finite continuations, and Theorem 5.6 gives exactly two complete status classes.

For a concrete instance take $X = \{p_1, p_2, n\}$, $X_+ = \{p_1, p_2\}$, and $X_- = \{n\}$. Let the only primitive observation be qualification. Let one total operation exchange p_1, p_2 and fix n ; let another be the identity restricted to X_+ . Domains and output bits are constant on both fibres. Every finite word therefore has the same definedness and final qualification at p_1, p_2 . If the two operations have warranted edges, the greatest safe interior is X_+ , and its quotient has one positive class. This proves the desired binary form in an actual inhabited model without assuming that arbitrary richer observations also disappear.

If the intended application needs to distinguish p_1 from p_2 , add that required observation. The complete profile then separates them. The original qualification remains binary. Completeness determines which of these interfaces serves the task; the theorem does not authorize choosing the less informative one merely because it has fewer states.

9.3 Simply typed construction: an independently justified interior

Consider the usual simply typed lambda calculus with Boolean constants and conditionals. Types and terms are

$$\begin{aligned} T &::= \text{Bool} \mid T \rightarrow T, \\ t &::= x \mid \text{true} \mid \text{false} \mid \lambda x : T. t \mid tt \mid \text{if } t \text{ then } t \text{ else } t. \end{aligned}$$

Use the ordinary variable, abstraction, application, Boolean, and conditional typing rules. In particular,

$$\frac{\Gamma \vdash t : T \rightarrow U \quad \Gamma \vdash u : T}{\Gamma \vdash tu : U}, \quad \frac{\Gamma \vdash c : \text{Bool} \quad \Gamma \vdash u : T \quad \Gamma \vdash v : T}{\Gamma \vdash \text{if } c \text{ then } u \text{ else } v : T}.$$

Let X consist of closed raw terms, modulo capture-avoiding renaming of bound variables. The reduction relation is fixed syntactically on raw terms before qualification is defined. Use the usual call-by-value rules: reduce the function position, then the argument position of an application with value function; reduce the guard of a conditional; contract a beta-redex with a value argument; and select the appropriate branch when the guard is **true** or **false**. Values are the two Booleans and abstractions. Fix a target type T and put

$$A_T = \{t \in X : \emptyset \vdash t : T\}.$$

The standard preservation theorem states that $t \in A_T$ and $t \rightarrow t'$ imply $t' \in A_T$. Its key beta step is the substitution lemma; congruence steps use the induction hypothesis, and Boolean selection uses the branch typing premise. Progress additionally says that a closed typed term is a value or has a step [6, 7]. Neither statement says that every raw predecessor of a typed term was typed.

Proposition 9.1 (The complete typed interior). *For this raw reduction relation, true operational edge warrants, and local qualification A_T , the greatest safe interior is exactly A_T . It is the unique complete reduction-stable qualification set, although it generally has many inequivalent members.*

Proof. Preservation proves the right-hand side of Theorem 6.7; hence $S_T = A_T$. Existence and uniqueness of the complete interior follow from Theorem 6.3 and Corollary 6.5. Membership is nonvacuous at $T = \text{Bool}$ because the typing rules derive both Boolean constants. Nothing in greatestness identifies those terms. $\square$

This is a substantive positive realization: reductions were not defined to be only the ones that preserve a set chosen afterward. The syntactic reduction rules are independent; the preservation theorem proves their behavior. If nonvalue states are required to have a step, progress supplies that readiness condition on A_T . No normalization or liveness claim is needed for the finite-path safety result.

Tests of proposed typed interfaces. At $T = \text{Bool}$, let b_T be the characteristic function of A_T . Both `true` and `false` have $b_T = 1$, while the original observation “is this the value `true`?” has values 1 and 0. They are an explicit fibre collision for a proposed bit-only value interface, under Proposition 8.2. If a declared partial operation consumes only the value `true`, the same pair also violates the primitive domain-saturation condition in Theorem 5.8. The typed objects can retain the necessary information; their common typing answer remains binary.

A proposed complete typed interior $J = \{\text{true}\}$ also fails a concrete test. It is qualified and reduction-closed because the value has no outgoing reduction. But `false` $\in A_{\text{Bool}} = S_{\text{Bool}}$ is omitted. The singleton $\{\text{false}\}$ is another sound invariant region, so Theorem 8.3 supplies an explicit completeness witness. Restricting attention to J can be a sound partial practice; claiming that it exhausts the typed interior is false. This distinction uses the positive preservation proof above.

There is also an explicit negative-to-positive raw step:

$$\text{if true then true else (true true)} \longrightarrow \text{true}.$$

Call the left term u . It is untypable: typing its unused branch would require the Boolean constant to have an arrow type. The right term has type `Bool`. A proposed checker that certifies original Boolean typing whenever it finds a Boolean-typed reduct therefore returns 1 at u , although its original typing has $b_T(u) = 0$. With

$$G_T = \{(v, b_T(v)) : v \in X\},$$

its asserted record $(u, 1)$ lies outside G_T , whose original record at u is $(u, 0)$. This is a surplus witness for Theorem 8.1; the displayed reduction does not justify the converse of preservation. The checker has enough syntax to perform the reduction but applies an invalid implication, so this error differs from the information loss in the preceding fibre collision.

The same step shows that raw reduction does not make untypability absorbing. The original term u remains untypable; the new right-hand term is a different term with its own derivation. Preservation, safe-interior uniqueness, and original-term conservation all remain intact.

Finally, syntax is already structure. A parser or type checker may use the constructor tree, annotations, and variable binding to determine the typing judgment. Such use does not violate the equality-only proposal theorem, whose premise is absent once those structures are licensed. If the whole parsed term is replaced by an opaque unmarked name, its typing cannot in general be recovered from that name alone. These two representations must not be assigned the same informational content merely because both are called proposals.

9.4 An original answer can be unknown and still conserved

Suppose a trial record is sealed. One investigator reports “unresolved”; another later opens a reliable record and correctly reports that the original unsupported trial passed. The earlier nonanswer and later positive answer do not disagree on a common answered domain. Proposition 5.1 states

exactly this domain-sensitive conservation. If the earlier investigator instead asserted failure without warrant, replacing that report corrects an error. It does not turn two discordant assertions into two individually warranted answers.

The same distinction appears in proof verification. A previously unread derivation can be valid under the original rules. A later check can establish that fact. By contrast, inserting a missing premise creates a different proof performance. Neither case supports a ban on new evidence or a claim that the correct original answer depends on whether it had been checked at the time.

10 The canonical architecture and its downstream use

The preceding results are connected by more than an analogy. When a complete interface retains the primitive operations and their warrant conditions, forming the admissible interior commutes with passage to that interface. This supplies an exact bridge between classification and standing.

10.1 The complete quotient preserves the complete interior

Consider the partial-operation system of Theorem 5.9. Its underlying configuration set is $E = X$. For this theorem the declared relation is exactly the graph family of the primitive operations:

$$R = \{(x, a, t_a(x)) : a \in L, x \in \text{dom}(t_a)\},$$

with the same local predicate A and edge-warrant predicate W used to define S . Thus there are no additional transitions hidden from the primitive profile. Include among the primitive Boolean observations the local qualification predicate A and, for every operation label a , the predicate

$$w_a(x) = 1 \iff t_a(x) \text{ is defined and } W(x, a, t_a(x)).$$

Let $\equiv$ be equality of the resulting finite-continuation profile. Theorem 5.9 supplies induced partial operations $\bar{t}_a$ on $\bar{X} = X/\equiv$. The empty-word observations make $\bar{A}([x]) = A(x)$ and the edge-warrant truth values well-defined. Set

$$\bar{R} = \{(z, a, \bar{t}_a(z)) : a \in L, z \in \text{dom}(\bar{t}_a)\}.$$

Let $\bar{S}$ be the greatest safe interior of this quotient system.

Theorem 10.1 (Canonical quotient–interior compatibility). *With these data,*

$$S = q^{-1}(\bar{S}), \quad q(S) = \bar{S}, \quad S/\equiv \cong \bar{S},$$

where $q : X \rightarrow \bar{X}$ is the quotient map and the last equivalence is the canonical bijection. Thus the unique complete stable interior can be formed before or after passing to the complete continuation interface, with exactly the same represented result.

Proof. Each original path maps to a quotient path with the same qualification and edge-warrant values. Conversely, a quotient path beginning at $[x]$ can be lifted beginning at that particular representative x . At each step, domain saturation ensures that the named primitive operation is defined at the current representative; its output represents the indicated next class by partial descent. Finite induction lifts the whole path. The observations A, w_a preserve every safety condition during both procedures. Hence all original paths from x are safe exactly when all quotient paths from $[x]$ are safe. This proves $S = q^{-1}(\bar{S})$. Surjectivity of q then gives $q(S) = \bar{S}$, and equality of its restricted fibres gives the canonical bijection. $\square$

This theorem explains why operation domains and edge warrants belong in a complete interface. Retaining only immediate truth values would not justify lifting a quotient path from every representative. The theorem establishes lossless transfer for interfaces satisfying these domain and warrant conditions.

Corollary 10.2 (Two-state specialization). *If the primitive descent conditions of Theorem 5.8 hold for a qualification bit b , including the local qualification and edge-warrant observations, the entire finite-context system descends to the attained bit space $b(X)$. Its complete stable interior is represented there without loss. If both bit values occur and the bit alone preserves every required observation, the complete interface has exactly two classes. If $A = b^{-1}(1)$ and this positive class is closed with warranted edges, stable standing equals that class.*

Proof. Primitive descent generates all finite-context observations. Theorem 5.6 identifies their quotient with $b(X)$ when those observations cover the intended family. Theorem 10.1 transports the greatest interior, and Theorem 6.7 gives the final equality. $\square$

The result is universal over systems satisfying the exhibited primitive checks; it is not a universal assertion that the checks succeed for every richer domain. Where original bearer data are retained separately, Corollary 5.7 gives the stronger appropriate relative statement: binary qualification can accompany a rich determinate object without replacing its full identity and behavior.

10.2 One theorem chain, with distinct sources of necessity

Theorem 10.3 (Canonical admissibility architecture). *Every complete realization of the stated determinate-use problem has the following canonical structures.*

- (a) *Its identity-preserving reference information factors through the complete reference quotient.*
- (b) *Every scalar answer claimed to be recoverable from a proposed lower interface passes the fibre criterion. Equality-only pre-fixational content imposes the exact equality-pattern and selector restrictions of Section 3.*
- (c) *Its complete observation frame determines a unique minimal lossless interface up to the unique commuting bijection. Its complete single-question qualification is binary; binary compression of the whole frame is characterized by Theorem 5.6.*
- (d) *Its independently declared safety/reuse contract determines a unique complete interior S . Under the complete primitive interface described above, that interior is invariant under quotienting and faithful transport.*
- (e) *Against the independently complete original use semantics, every failed claim of complete realization has a surplus, omission, or warranted-use defect. Theorem 8.6 locates these defects in the five-locus account. The absence of a proof of coverage is not itself a semantic counterexample, but prevents an unconditional claim that coverage has been established.*
- (f) *Faithful answers about one original incidence agree, and retained histories cannot erase an earlier failed qualification. The additional endpoint claim is valid exactly under the criterion in Theorem 7.2.*

All successful determinate uses of these structures instantiate the kernel roles necessarily.

Proof. Part (a) is Theorem 2.4. Part (b) combines Theorem 4.1 with Theorem C.9; their different hypotheses are explicit. Part (c) is Theorem 5.5 and its binary specialization. Part (d) is Theorem 6.3, Corollary 6.5, and Theorems 6.8 and 10.1. Part (e) is Theorem 8.6, with its independent coverage obligation. Part (f) follows from Theorem 5.2 for arbitrary partial answers, Proposition 5.1 for its Boolean specialization, and Theorem 7.2 for retained paths. Proposition 2.3 supplies the final constitutive conclusion. None of the parts imports the completed structure theorem as a premise. $\square$

Three kinds of necessity occur here. Kernel necessity concerns the constitutive work of determinate use. Canonicity concerns mathematical structures uniquely specified by complete data. Particular closure or symmetry conditions concern substantive domain facts. Their distinction is productive: it tells an application exactly what is unavoidable, what is unique once the data are fixed, and what still needs a domain proof.

10.3 Mathematics and physics

In mathematics, the theorem can organize judgments, derived rules, reduction systems, certificate interfaces, and constructions defined only on particular types or contexts. The domain proof must establish the relevant original inference conditions and operation behavior. A quotient that forgets proof identity may still preserve provability; a quotient preserving provability may fail to preserve normalization observations or a required application domain. The complete question determines the obligation.

In physics, the same roles are incurred by the actual determinate incidence, including unobserved events and records. A model supplies claims about its carrier, conditions, transformation laws, and physically relevant comparisons. Applying the architecture requires proving the covariance or invariance claimed for those data. It cannot be established by replacing the physical symmetry group with the full symmetric group of a set or by treating a dimensionless variable as structureless. Conversely, an established determinate physical incidence does not wait for a separate permission to be kernel-governed.

A proposed downstream theorem should state the original question and bearer, specify the licensed information, verify the claimed transformations, retain the full required observation family, and establish the continuation conditions. If it claims a unique complete interior, it must prove coverage as well as exclusion. If it claims one physical or mathematical occupant, rather than one complete set or quotient, it needs a further inhabitation-and-uniqueness proof in the relevant fibre. The existence of a unique admissible *set* does not make that set a singleton.

10.4 Assessment of the theorem's force

The completed structure supplies positive constructions and exact obstructions. It gives a quotient with a universal property, a stable interior with a greatestness proof, finite-path witnesses to failure, and a transport theorem relating the two. The exhaustion theorem makes the negative side of that construction explicit: excess has an unsupported-result or unsafe-use witness, omission has a missed original result or safe configuration, and defects in the claimed warrant cannot be repaired by presentation alone. It also gives exact criteria for the stronger binary and endpoint forms when their local hypotheses are established. These results do not classify mathematics, physics, or reality as one closed field of objects; they classify the admissibility architecture of the complete determinate-use problem.

The live challenges are therefore concrete. A purportedly complete interface can be defeated by a missed observation. A symmetry claim can be defeated by omitted licensed structure. An exhaustion claim can be defeated by an omitted original obligation or a failed correspondence between its witness and the asserted disposition. An interior claimed complete can be defeated by a sound closed component it omits. A proposed conservation law can be defeated by discordant individually warranted answers for one complete original incidence. A proof of authorization can be challenged at its semantic premises or rule soundness. Each is a substantive point of contact with a realization.

What remains universal is the dependence emphasized at the beginning: determinacy, identity, comparison, admissible transformation, standing-bearing use, and original fixation incur the kernel work. The structure theorem makes the interfaces of that work explicit, with exact conditions for its use in further constructions.

A Claim map and proof dependencies

This appendix records the exact strengths of the reference theorem chain. It also identifies where a broader formulation would require an additional argument. The purpose is to make later citations preserve the proved subject and scope.

Kernel necessity.

Proposition 2.3: necessary functional work of successful determinate use. The fuller constitutive reconstruction is in the Kernel reference manuscript. Particular physical laws and qualification predicates remain substantive data.

Reference stabilization.

Theorem 2.4: the quotient is a complete invariant; nonconstant separation requires at least two classes. Nonemptiness of the family of invariant maps alone is insufficient.

AMetricity.

Theorem C.9: equality-pattern classification under full equality-only content. Theorem C.6 handles the actual symmetry group for richer content. Dimensionlessness alone does not establish full permutation symmetry.

Boundary non-transmission.

Theorem 4.1 and Corollary 4.2: exact answer descent iff fibre constancy. A missing distinction cannot be recovered by postprocessing; a justified added channel can supply it.

No authority bootstrap.

Theorems 4.4 and 4.8: locate semantic warrant and exclude a wholly unseeded rule closure. They do not prohibit new proofs, implementations, or sufficient certificates.

Bivalent qualification.

Proposition 5.1: each complete classical qualification has a binary truth value; individually sound reports agree on their common domain. Nonanswers and evidence states are separate.

Complete status quotient.

Theorem 5.5: exact canonical quotient of all required observations. Theorem 5.6 gives binary reduction iff all observations are constant on qualification fibres; exactly two classes additionally requires both bits.

Compositional completeness.

Theorems 5.8 and 5.9: primitive observations, domains, and output classes generate all finite continuations. Semantic coverage of the intended uses must be established independently.

Standing identity.

Theorem 6.3: qualification for the robust-reuse question is the greatest safe interior. Theorem 6.7: it equals immediate admission exactly when local preservation and edge warrants hold.

Unique interior.

Corollary 6.5: exactly one complete or inclusion-maximal sound inductive candidate. Arbitrary proper sound subsets can differ. The interior can have many elements and many reference classes.

Conservation.

Theorem 5.2: collective original-incidence agreement, including partial outputs and definedness. Corollary 5.3: exact partial equivariance. Proposition 5.1: the Boolean specialization. Proposition 7.1: positive safe continuation. Theorem 7.2: retained-history non-repair and exact endpoint criterion. Negative absorption is not inferred from historical fixation alone.

Exhaustion and impossibility.

Theorems 8.1, 8.3, and 8.6: exhaustive semantic discrepancies, finite unsafe-use witnesses, and their failure-locus coverage. Completeness concerns the independently specified original task. Missing evidence is not a false answer; partial sound certification is not a failed claim of completeness; the five mechanisms need not be disjoint or irreducible.

Canonical transport.

Theorems 6.8 and 10.1: a faithful relation isomorphism or complete primitive quotient transports the unique complete interior. Coverage of target continuations is essential.

A.1 Acyclic proof order

The constitutive kernel argument fixes the target work. The reference quotient, equality-pattern theorem, stabilizer criterion, and fibre theorem have direct local proofs. The semantic authorization argument and rule-closure obstruction depend on their explicit interpretations. The complete observation quotient is constructed from the original required questions, without assuming binary compression. Primitive descent proves the finite-context consequences. The stable interior is then constructed from independently specified local and edge conditions, not imported from a uniqueness assertion. Conservation uses the original incidences. Exhaustion compares claimed graphs and interiors with their independent semantic targets and proves the failure-locus correspondence; it does not define the target by those loci. Quotient–interior compatibility uses the independently proved descent and path criteria.

This acyclic order provides the proof dependencies for subsequent applications. The AMetric reference manuscript develops the broader boundary mathematics; Appendix C provides the full local results used here.

A.2 Logical and computational conventions

The mathematical proofs use classical set theory. Bivalence here is the characteristic predicate of a complete qualification proposition in that metalogic. The manuscript does not purport to derive

excluded middle for every alternative logic or settle every incomplete physical question. No set-sized family is assumed finite unless stated. An exact quotient or greatest fixed point need not have a computable presentation.

The stabilizer existence converse states where ordinary choice is used; its uniqueness form does not require orbitwise choices. The interior proof quantifies over finite paths and does not require a chosen infinite branch. The finite-context proof requires a specified grammar and an independently justified coverage claim. Safety does not imply fairness, termination, controllability, or satisfaction of an eventual-outcome condition. If a domain requires such properties, they must be incorporated through their own semantics and proofs; they are not obtained by calling safety “complete.”

These conventions concern mathematical realization. They do not make kernel governance optional for determinate objects in a different notation, logic, or physical theory. What changes is which particular classical representation theorem has been established for that realization.

B Notation, scope, and verification sequence

Symbol	Meaning in its local section
D	Original assessment contract: bearer, question, conditions, and intended uses.
X/E	Reference quotient in Section 2.2; there E is an equivalence relation.
$P, \text{Sym}(P)$	Equality-only proposal carrier and its full permutation group in Section 3.
E, p, a	Complete candidate situations, retained description, and required target answer in Section 4.
$b, o_q, P(e)$	Qualification bit, observation coordinates, and complete profile in Section 5. The profile notation is local to that section and its consequences.
$t_a, d_w, v_{w,j}$	Primitive partial operations, word-definedness, and final Boolean observations.
X, L, R, A, W	Configurations, operation labels, declared transitions, local qualification, and edge warrants.
Φ, S, U_n	Reuse operator, greatest safe interior, and descending finite-depth approximants.
H	All-stages qualification of a retained finite history.

B.1 Four quantities that must not be identified

1. *A truth value for one complete use.* This answers the fixed qualification proposition. It is not the complete identity of the bearer.
2. *The complete use profile.* This includes relevant definedness and contextual answers. It is the subject of the lossless-interface theorem.
3. *The complete stable interior.* This is a subset of configurations satisfying the robust-reuse question. It is unique as a complete subset, not necessarily a singleton or one orbit.
4. *The retained historical claim.* This concerns original incidences and cannot be repaired by replacing them with new successful incidences. It is different from current endpoint qualification.

The bridge and typed examples supply witnesses to each difference. A positive split in a complete profile is a legitimate objection to a bit-only encoding. An omitted closed component is a legitimate objection to claimed interior completeness. A new qualified endpoint is compatible with retained failure in its history. These observations are structural tests, not exceptions to necessary governance.

B.2 Recommended verification sequence

First verify the independently specified original bearer and complete question. Next inspect the content inventory behind the symmetry claim: a hidden relation can invalidate full permutation symmetry even without selecting an atom. Apply the fibre criterion to the proposed reduction, including undefinedness and output interpretation. Check each primitive continuation for domain saturation and output descent, then verify that the generated contexts cover the intended uses. For a proposed interior, establish local qualification, edge warrants, closure, and completeness separately. For exhaustion, inspect how the required observations and uses were obtained from the original task, then verify the surplus/omission witnesses and their correspondence to the named failure mechanisms. Missing proof of coverage must be reported as an unresolved obligation rather than as a semantic negative. Finally identify which conservation statement is claimed and whether its subject is the original incidence, a new configuration, or a retained path.

This sequence also locates errors in implementation. A decoder may be correct on its attained image but undefined elsewhere; a checker may be sound but incomplete; a transport may preserve old edges while introducing new unsafe target edges; and a finite test may check examples without establishing a general theorem. Each deserves its own statement rather than an undifferentiated claim that the interface has been verified.

B.3 Formalization boundary

The theorem chain is established by the mathematical proofs in this paper, with explicit finite-model verification in the accompanying project. The complete chain has not been checked by a proof assistant. Existing kernel formalizations provide separately scoped evidence for their stated results. Formalizing the quotient, fixed-point, and transport theorems requires preserving the independently declared observation family, operation domains, local qualification, and edge warrants, and exposing the assumptions of each exported theorem.

C Supporting AMetric classification and selection proofs

This appendix supplies the full proofs supporting Section 3. The classification, selection, metric, and probability results give a self-contained treatment of the local boundary. The boundary concerns what may be used to determine a target before the relevant determining structure has been supplied. It is relative to that target role. A proposal record is already a determinate record, and hence already kernel-governed, even when it supplies no warrant for the proposed target incidence. Nothing here postulates an ungoverned physical realm beneath determinate objects.

C.1 From absence of an individuator to the right symmetry

Definition C.1 (Licensed proposal content). For a proposed target use, the licensed proposal content is the complete structure that may bear on that use before the proposed fixation. Its automorphisms preserve all that content. An *equality-only carrier* is a set P whose licensed structure for this role consists solely of equality, with no designated atom, order, coordinate, adjacency, probability weight, or other target-bearing relation. Its automorphism group is $\text{Sym}(P)$.

An absence claim requires an inventory of the content actually licensed. Dimensionlessness rules out an already given dimensional scale; it does not rule out a dimensionless ratio, a graph relation, or a distinguished parameter. The further absence of a licensed individuator rules out a justified choice of one candidate. It does not by itself rule out relations among candidates. An unmarked finite cycle, for example, has no vertex fixed by all rotations while retaining adjacency. Its automorphism group is smaller than the full permutation group when the cycle has at least four vertices.

The analysis has three steps. First identify the target-bearing content. Next identify its actual automorphisms. Then ask what predicates or selections that symmetry permits. When the content is genuinely equality-only, every permutation preserves it and the following complete classification applies. When more content is present, the same stabilizer and fibre tests still apply, using the smaller justified symmetry group. This is a local mathematical distinction within necessary kernel governance.

Proposition C.2 (Pre-fixational anti-haecceity). *Let G be the automorphism group of the full licensed proposal content. A target-bearing predicate on proposals that is warranted solely by that content must be G -invariant. A deterministic construction determined by this licensed content, with a structured output, must be equivariant under the output action independently justified by its target interpretation. That action is not selected retrospectively to make a preferred construction equivariant. A dependence on labels not retained by the licensed content cannot supply target authority from that content alone.*

Proof. An automorphism changes presentation while retaining every licensed relation on which the target inference may depend. A purported inference from that content alone must give the same claim after the corresponding identification of inputs and outputs. If it does not, its rule uses a distinction the preserved content does not supply, or the two outputs fail to express the same target inference. In the first case the advertised content inventory is incomplete; in the second the rule is not a faithful inference from it. For a scalar truth value the output action is trivial, giving invariance. For a structured output the corresponding condition is equivariance. $\square$

The proposition connects faithful use of the declared content to equivariance. Set theory supplies many non-equivariant functions on a named set. The question is whether one is warranted by the declared content for the stated target. A numeral printed beside an atom is additional content if that numeral is to choose it.

C.2 Complete finite-arity classification

For $\bar{p} = (p_1, \dots, p_n) \in P^n$, write $\text{Eq}(\bar{p})$ for the partition of $\{1, \dots, n\}$ defined by $i \equiv j$ exactly when $p_i = p_j$. Only partitions with at most $|P|$ blocks are realized. The action of $\text{Sym}(P)$ on tuples is diagonal.

Lemma C.3 (Equality-pattern orbits). *Two finite tuples in P^n lie in the same permutation orbit if and only if they have the same equality pattern.*

Proof. A permutation preserves equality. Conversely, if the patterns agree, the assignment $p_i \mapsto q_i$ is a well-defined bijection between the two finite sets of entries. Complete this finite partial bijection to a permutation of their union, and fix every other element of P . The resulting permutation sends one tuple to the other. This also treats the empty tuple. $\square$

Theorem C.4 (Exact equality-only predicate theorem). *For a fixed carrier P and finite arity n , a relation $R \subseteq P^n$ is invariant under all permutations exactly when it is definable on P by a parameter-free quantifier-free formula using only equality. Equivalently, it is a union of realized equality-pattern classes. Every invariant scalar map $f : P^n \rightarrow Z$, with trivial action on Z , factors uniquely through the equality-pattern map onto its image.*

Proof. The orbit lemma identifies the classes on which an invariant relation or scalar map is constant. For a partition π of $\{1, \dots, n\}$, let

$$\theta_\pi(\bar{x}) = \bigwedge_{i \sim_\pi j} (x_i = x_j) \wedge \bigwedge_{i \not\sim_\pi j} (x_i \neq x_j).$$

There are finitely many partitions. The disjunction of θ_π over the realized classes belonging to R defines R ; an empty disjunction is false. Every equality formula is invariant, since equality and Boolean combinations are invariant. The scalar factorization assigns to a realized pattern the common value of its tuples; well-definedness and uniqueness follow from orbit constancy and surjectivity onto realized patterns. $\square$

The theorem is stronger than a claim that expressions in an equality language happen to be invariant: at each fixed finite arity it characterizes *all* invariant predicates. It does not assert that one formula works uniformly across all possible cardinalities. Nor does it erase the finite incidence structure of a tuple. The relation $x_1 = x_2$ is informative about a pair, even though no unary relation can distinguish unmarked atoms of a nonempty carrier.

Corollary C.5 (Unary constancy and order obstruction). *On a nonempty equality-only carrier, an invariant unary predicate is either everywhere true or everywhere false. An invariant strict asymmetric relation on the atoms has no related pair. In particular, no nontrivial strict ranking is supplied by the unmarked atom structure.*

Proof. There is one unary equality pattern. For a strict relation there are no diagonal pairs; if $p \prec q$ for $p \neq q$, transposing p, q gives $q \prec p$, contrary to asymmetry. $\square$

The empty strict relation survives. A constant score and a universally accepting or rejecting classifier also survive. These facts matter: AMetricity excludes unsupported discrimination, not every formally definable object.

C.3 The stabilizer test and what selection requires

Let a group G act on input space Z and output space Y . Write $G_z = \{g : gz = z\}$ and $\text{Fix}(G_z) = \{y : gy = y \text{ for all } g \in G_z\}$. Suppose $B_z \subseteq Y$ is the independently specified set of eligible outputs, transported by $B_{gz} = gB_z$.

Theorem C.6 (Eligible stabilizer criterion). *An equivariant selection f with $f(z) \in B_z$ can exist only if*

$$B_z \cap \text{Fix}(G_z) \neq \emptyset \quad (z \in Z).$$

Conversely, under ordinary set-theoretic choice, this condition for one representative of every orbit suffices. If each such intersection is a singleton, an equivariant selection exists uniquely without choosing orbit representatives.

Proof. If $h \in G_z$, then $hf(z) = f(hz) = f(z)$, so the selected output is eligible and fixed. Conversely choose one representative z_O and one $y_O \in B_{z_O} \cap \text{Fix}(G_{z_O})$ for each orbit. Set $f(gz_O) = gy_O$. If $gz_O = g'z_O$, then $g'^{-1}g \in G_{z_O}$ fixes y_O , making the definition independent of the presentation. Eligibility and equivariance follow from transport. In the singleton case define $f(z)$ directly as the unique member of $B_z \cap \text{Fix}(G_z)$; conjugacy of stabilizers transports these singletons and proves equivariance and uniqueness. $\square$

Corollary C.7 (No unmarked atom selection). *If $|P| \geq 2$, there is no $\text{Sym}(P)$ -equivariant selection of one element of P from an unmarked input with trivial group action. There is, however, a canonical orbit or equality-class code; it does not select an atom.*

Proof. The stabilizer of the unmarked input is all of $\text{Sym}(P)$, which fixes no atom when $|P| \geq 2$. The quotient code refers to the orbit itself, which is invariant. $\square$

A supplied point, ordering, or sufficiently rich structural witness can make an eligible fixed point available. For instance, a tuple supplies positions and entries; projecting to its first entry is equivariant. A marked vertex on a cycle changes the stabilizer. Selection after such additions is a lawful realization if the additions are licensed for the target. Calling their use “merely a representation” does not eliminate them from the data that explain the choice.

C.4 Metrics, probabilities, and typicality

Proposition C.8 (Exact invariant atom metrics). *For $|P| \geq 2$, every real-valued metric on P invariant under all permutations has the form*

$$d(p, q) = \begin{cases} 0, & p = q, \\ c, & p \neq q, \end{cases} \quad c > 0.$$

Every such function is a metric. It supplies no ranking among distinct unmarked atoms and no internally preferred positive scale.

Proof. The off-diagonal pairs form one orbit, so their common distance is a constant $c > 0$. The metric axioms hold for every positive c : the only nonzero triangle inequality required is $c \leq c + c$, or $c \leq c$ when two vertices coincide. No atom or distinct pair receives a privileged distance. $\square$

The discrete metric is thus a real mathematical survivor. Hamming metrics on tuples can likewise depend on supplied coordinate comparisons. The AMetric conclusion concerns the absence of a distinguished magnitude or target-bearing discrimination furnished by the bare carrier. A convention choosing $c = 1$ is usable as a convention; it does not recover a missing physical scale from pure equality.

On a finite carrier of size $m > 0$, the uniform distribution assigning $1/m$ to each atom is permutation-invariant. It supplies a distribution, not an equivariant deterministic choice of its realized outcome. On a countably infinite carrier, a countably additive probability assigning the same mass to every singleton cannot total one: a positive mass diverges under summation, while zero masses sum to zero. These are different cases. Randomization can be part of an authorized physical or computational procedure, but it does not retrospectively turn an unmarked input into a content-determined selected atom.

A unary “typical” class defined solely from equality is empty or universal by Corollary C.5. Typicality using a measure, topology, or asymptotic family can be substantive, because those are additional licensed structures. Its use requires that the additional structure be present and justified for the target.

C.5 The local boundary conclusion

Theorem C.9 (AMetric proposal-side constraint). *A target use whose complete licensed proposal content is equality-only admits exactly the finite-arity invariant predicates classified by Theorem C.4. It cannot obtain nontrivial unary discrimination from that content alone; when $|P| \geq 2$, it cannot obtain an unmarked atom choice either. A proposed finer determination must either supply further licensed structure or fail to be determined by the advertised input. A construction using a smaller, justified symmetry group is governed by its eligible stabilizers, not by an unjustified full-permutation assumption.*

Proof. Proposition C.2 connects faithful use to invariance or equivariance. The equality-pattern theorem, unary corollary, and stabilizer criterion then apply. An output that separates proposals within an orbit cannot be invariant under the retained content, so it uses additional information or is not a faithful determination from that content. The same argument works with the actual automorphism group of richer content. $\square$

The boundary becomes a non-transmission theorem when the exact target answer is specified. That next step is important: a classification of invariant proposal structure is not yet a classification of every evaluated state. Evaluation may introduce a licensed measurement, premise, relation, or use-condition. Section 4 identifies exactly when such information has been retained and where a successful certificate obtains its warrant.

References

- [1] Wiggins, D. (2001). *Sameness and Substance Renewed*. Cambridge University Press. <https://doi.org/10.1017/CB09780511612756>.
- [2] Stroud, B. (1968). Transcendental arguments. *The Journal of Philosophy* **65**(9), 241–256. <https://www.jstor.org/stable/2024395>.
- [3] Pitts, A. M. (2013). *Nominal Sets: Names and Symmetry in Computer Science*. Cambridge Tracts in Theoretical Computer Science, vol. 57. Cambridge University Press. <https://doi.org/10.1017/CB09781139084673>.
- [4] Rutten, J. J. M. M. (2000). Universal coalgebra: a theory of systems. *Theoretical Computer Science* **249**(1), 3–80. [https://doi.org/10.1016/S0304-3975\(00\)00056-6](https://doi.org/10.1016/S0304-3975(00)00056-6). Author-institution record: <https://ir.cwi.nl/pub/48>.

- [5] Tarski, A. (1955). A lattice-theoretical fixpoint theorem and its applications. *Pacific Journal of Mathematics* **5**(2), 285–309. <https://msp.org/pjm/1955/5-2/pjm-v5-n2-p11-s.pdf>.
- [6] Pierce, B. C. (2002). *Types and Programming Languages*. MIT Press.
- [7] Pierce, B. C., and collaborators. *Software Foundations*, vol. 2: *Programming Language Foundations*, chapter “Properties of STLC.” Online edition, accessed 20 September 2026. <https://softwarefoundations.cis.upenn.edu/plf-current/StlcProp.html>.
- [8] Maley, A. J. (2026). *Non-Degenerate Construction and the Kernel of Admissibility*. Reference manuscript, version 49. Unpublished manuscript.
- [9] Maley, A. J. (2026). *The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction: Pre-fixational authority, non-transmissivity, and complete interfaces*. Editorial reference edition. Unpublished manuscript.